Applied Energy 369 (2024) 123541

Contents lists available at ScienceDirect

Applied Energy

journal homepage: www.elsevier.com/locate/apenergy

Harnessing AI for solar energy: Emergence of transformer models

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HIGHLIGHTS

- Addressing the critical need for accurate forecasting of solar irradiance and photovoltaic generation.
- Examining transformer advancements in solar forecasting, critical for integrating solar energy into power grids.
- Analyzing model capabilities in diverse environments, highlighting their role in forecasting accuracy and deployment issues.
- Identifying future research in model standardization and computational efficiency to improve solar forecasting.

ARTICLE INFO

Keywords:
Transformer models
Machine learning (ML)
Deep learning (DL)
Solar energy forecasting
Photovoltaic (PV)
Renewable energy integration

ABSTRACT

This review emphasizes the critical need for accurate integration of solar energy into power grids. It meticulously examines the advancements in transformer models for solar forecasting, representing a confluence of renewable energy research and cutting-edge machine learning. It evaluates the effectiveness of various transformer architectures, including single, hybrid, and specialized models, across different forecasting horizons, from short to medium term. This review unveils substantial improvements in forecasting accuracy and computational efficiency, highlighting the models' proficiency in handling complex and diverse solar data. A key contribution is the emphasis on the crucial role of hyperparameters in refining model performance, balancing precision against computational demands. Importantly, the research also identifies critical challenges, such as the significant computational resources required and the need for expansive, high-quality datasets, which limit the broader application of these models. In response, this review advocates for future research directions focused on standardizing model configurations, venturing into longer-term forecasting, and fostering innovations to enhance computational economy. These proposed pathways aim to surmount current challenges, steering the domain towards more accurate, adaptable, and sustainable solar forecasting solutions that can contribute to achieving global renewable energy and climate objectives. This review not only maps the present landscape of transformer models in solar energy forecasting but also charts a trajectory for future advancements. It serves as a pivotal guide for researchers and practitioners, delineating the current advancements and future directions in navigating the complexities of solar data interpretation and forecasting, thereby significantly contributing to the development of reliable and efficient renewable energy systems.

1. Introduction

1.1. Background

The renewable energy sector, encompassing electricity, building energy, transport, and agriculture, is experiencing a surge in development, leading to increased demand and interest [1]. Among the various sources of renewable energy—solar, wind, hydropower, and geothermal—wind and solar power have gained significant traction and are now widely adopted across the globe [2,3]. Unlike traditional fossil fuels, renewable energy's reliance on weather conditions introduces a layer of variability, making the power supply from these sources more susceptible to climate-related changes. This variability presents a considerable challenge in maintaining a balance between the supply and demand of renewable energy [4].

To mitigate this issue, a substantial amount of research and development efforts has been directed towards Renewable Energy Forecasting (REF) techniques [5]. This focus has propelled REF to the forefront of

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https://doi.org/10.1016/j.apenergy.2024.123541
Received 5 February 2024; Received in revised form 3 May 2024; Accepted 22 May 2024
Available online 1 June 2024

scientific inquiry in the energy sector, marking it as one of the most vigorously pursued areas of research in recent years [6,7]. Through the advancement of REF methods, researchers and practitioners aim to enhance the predictability and reliability of renewable energy generation, thereby supporting the broader integration of renewable sources into the energy mix [8,9].The predictability of solar irradiance (SI) is crucial due to its sensitivity to a myriad of geographical and climatic factors. This variability directly impacts the stability and efficiency of photovoltaic (PV) integrated systems, as well as the broader energy market and the operational strategies of power utility agencies. Consequently, achieving precise and reliable forecasts of SI is essential for the seamless operation of the power grid [10].

1.2. Literature review

Over the past few decades, efforts to enhance REF have branched into two distinct but complementary directions: physical models and data-driven models [11,12]. Physical models leverage simulations based on physical laws, requiring the power of supercomputers equipped with heterogeneous hardware to process [13]. However, these models face computational challenges that have been difficult to surmount, prompting a shift towards data-driven models as a viable alternative [14].

Data-driven approaches, particularly favored in recent developments, encompass a variety of techniques. These include the Autoregressive Moving Average Model [15], Exponential Smoothing [16], and several Machine Learning (ML) strategies such as Support Vector Machine [17], Support Vector Regression, Kernel Extreme based Learning Machine [18], and Back Propagation Neural Networks [19]. Deep Learning (DL), an evolution of ML, has been recognized for its significant breakthroughs in addressing complex problems that have long stumped the artificial intelligence (AI) community [20]. Specifically, in the realm of REF, DL has demonstrated exceptional performance [21].

A common starting point for many DL approaches to REF has been the implementation of recurrent networks, notably Long Short-Term Memory Networks (LSTM) [22], Gate Recurrent Unit (GRU) [23], and Convolutional Long Short-Term Memory Networks (CNN-LSTM) [24]. Despite the preference for recurrent architectures in sequence modeling tasks, these networks are often criticized for their training complexities [25]. Conversely, research has indicated that convolutional architectures could serve as a viable alternative for REF, with studies demonstrating their effectiveness over recurrent networks in various applications [26,27]. However, convolutional networks face challenges in processing long sequences due to the need for a wide receptive field, which can increase the computational cost and potentially lead to training instability and performance degradation [28].

The exploration of ensemble learning and the fusion of recurrent and convolutional networks have yielded performance improvements in the short term [29,30]. Yet, these gains tend to plateau, and despite a few studies pushing the boundaries [31,32], the field of REF largely remains constrained to low temporal-resolution and short-term predictions, with minimal breakthroughs in extending forecasting horizons [33,34].

A series of innovative studies have introduced hybrid deep learning models to enhance solar radiation prediction accuracy. One approach combined convolutional networks for feature extraction with LSTM networks for prediction, demonstrating superior performance over traditional models [35]. Another study explored a CNN-LSTM methodology, analyzing data across 34 locations in diverse climate zones to refine prediction accuracy [36]. Further, a hybrid model employing LSTM and CNN, trained on meteorological data from various California locations, assessed the model's efficacy across different seasons and sky conditions [37]. Additionally, a novel deep learning framework was proposed, utilizing CEEMDAN for data decomposition from varied climate regions and designing CNN-LSTM structures for multiple feature inputs, showcasing higher prediction accuracy compared to other advanced models [38]. A unique hybrid deep-learning model was also introduced, merging machine learning techniques like WPD, CNN, LSTM, and MLP to predict solar radiation using both frequency and spatial features for the first time [39]. Moreover, an integrated model combining LASSO and LSTM highlighted the efficiency and accuracy of short-term solar radiation prediction, though it noted the potential computational and practical application challenges due to the model's complexity [40].

The introduction of the Transformer model has significantly advanced the field of sequence modeling, demonstrating distinct advantages over traditional recurrent neural network approaches [41]. Characterized by its streamlined architecture, the Transformer model excels at capturing global information, enhancing the overall learning efficiency through its capacity for parallel computing, are also coined as foundation models by some researchers [42]. This efficiency and structural simplicity have led to its successful application in various domains, including solar radiation prediction. Research leveraging the Transformer model for this purpose has reported noteworthy achievements, underscoring its potential to surpass existing methods in accuracy and computational effectiveness [43].

The Transformer model has revolutionized several domains within artificial intelligence (AI), including Computer Vision (CV), Automatic Speech Recognition, and Natural Language Processing (NLP), by offering groundbreaking approaches to complex problems [44]. Its versatility and effectiveness have led to its adoption across a wide range of AI tasks, positioning it as a pivotal technology in bridging the gaps between diverse research areas [45]. Its application holds the promise of significantly advancing REF, enabling the field to tackle more complex issues and contribute to societal progress [46]. Particularly in time series forecasting (TSF), Transformer-based models have garnered attention for their unique capability to facilitate direct information flow across vast sequences, enhancing the accuracy and efficiency of long-term predictions. Innovations such as the LogSparse Transformer, which offers reduced space complexity for extensive sequence forecasting [47], and the Informer, which improves upon the LogSparse Transformer with even lower complexity [48], exemplify the ongoing advancements within this space. Further developments include the Autoformer, which introduces an inner decomposition block to bypass traditional computation methods, achieving significant performance improvements [49], and the FEDformer, which sets new benchmarks in the field [50].

The integration of Transformer models into solar forecasting represents a significant leap forward. Their self-attention mechanisms adeptly manage sequential data, enabling selective focus on the most pertinent information within input sequences—a feature particularly advantageous for solar forecasting's dynamic requirements [51,52]. Recent explorations have demonstrated the superior capabilities of Transformer models, like the Temporal Fusion Transformer and the SL-Transformer, in outperforming conventional DL models and hybrid approaches in forecasting accuracy [53,54]. The superiority of transformer models in solar energy prediction stems not only from their architectural novelties but also from their ability to overcome the limitations present in other ML and DL frameworks [52,55]. For instance, transformer models are not as reliant on extensive temporal dependencies, which can be a constraint for models like LSTMs [53]. Furthermore, the flexibility and scalability of transformer models make them highly adaptable to a wide range of forecasting scenarios, from short-term to long-term predictions [54,56].

1.3. Contribution and configuration of the critique

This review critically assesses the role of transformer models in enhancing solar energy forecasting, juxtaposing them with other AI methodologies to offer a detailed comparative analysis. It endeavors to uncover the distinct advantages and untapped potential of transformer models, aiming to elucidate their significant contributions to the field of solar energy forecasting. The emphasis on these models is driven by their established effectiveness and their potential to navigate the complexities involved in accurately predicting solar energy outputs, a key factor in facilitating the widespread adoption of solar power.

The examination delves into the current landscape of solar forecasting, laying the groundwork for an in-depth investigation into how transformer models can push the boundaries of existing forecasting methodologies. This analysis is poised to enrich the corpus of renewable energy forecasting literature, spotlighting the pivotal role of cutting-edge DL technologies in bolstering the precision and dependability of solar energy forecasts.

Moreover, this review is methodically structured to cover various aspects of transformer models in solar forecasting. Section 2 introduces these models, dividing them into two primary categories: single model frameworks, focusing on individual transformer models, and hybrid model frameworks, which combine transformers with other computational methods for enhanced accuracy. Section 3 assesses the application of these models across different forecasting timeframes, examining their performance from short to long-term predictions. Section 4 sheds light on the computational efficiency of understudy Transformer models. Section 5 delves into the outcomes and key hyperparameters affecting these models, providing insights into their effectiveness and limitations. The review culminates with Section 6, discussing limitations and future research directions, followed by a conclusion. This structured approach aims to progressively deepen the reader's understanding, offering a thorough exploration of transformer models in the context of solar energy forecasting.

By merging the detailed examination of transformer models' current state and their structured critique, this review seeks to chart new territories in renewable energy research. It highlights the significant contribution of this comprehensive review to the existing body of research in renewable energy forecasting, aiming to not only elucidate the capabilities and advancements brought by transformer models but also to explore their future potential in revolutionizing solar energy predictions.

1.4. Bibliometric analysis

In an extensive review of transformer models implemented in solar forecasting, a comprehensive analysis of the scholarly landscape up until December 2023 reveals a total of 64 publications in this field, showcasing a diverse array of publication venues and a rich tapestry of research outputs (Fig. 1). At the forefront of this domain, IEEE has established itself as a leading disseminator of research, contributing 23 papers across various journals and conferences such as IEEE Transactions on Sustainable Energy and an impressive array of IEEE conferences. These include specialized events like the IEEE Energy Conversion Congress and Exposition and the IEEE International Conference on Big Data, signifying the organization's integral role in shaping the discourse on renewable energy technologies. Following closely, Elsevier's journals such as Energy, Applied Energy, Renewable Energy, and Solar Energy have collectively contributed 12 papers, anchoring the academic pursuit of sustainable energy practices within the scientific community. MDPI fortifies this collection with 11 papers from its own repertoire, including Energies and Applied Sciences, reflecting its commitment to multidisciplinary research in energy science.

The dynamic between different types of publications is equally

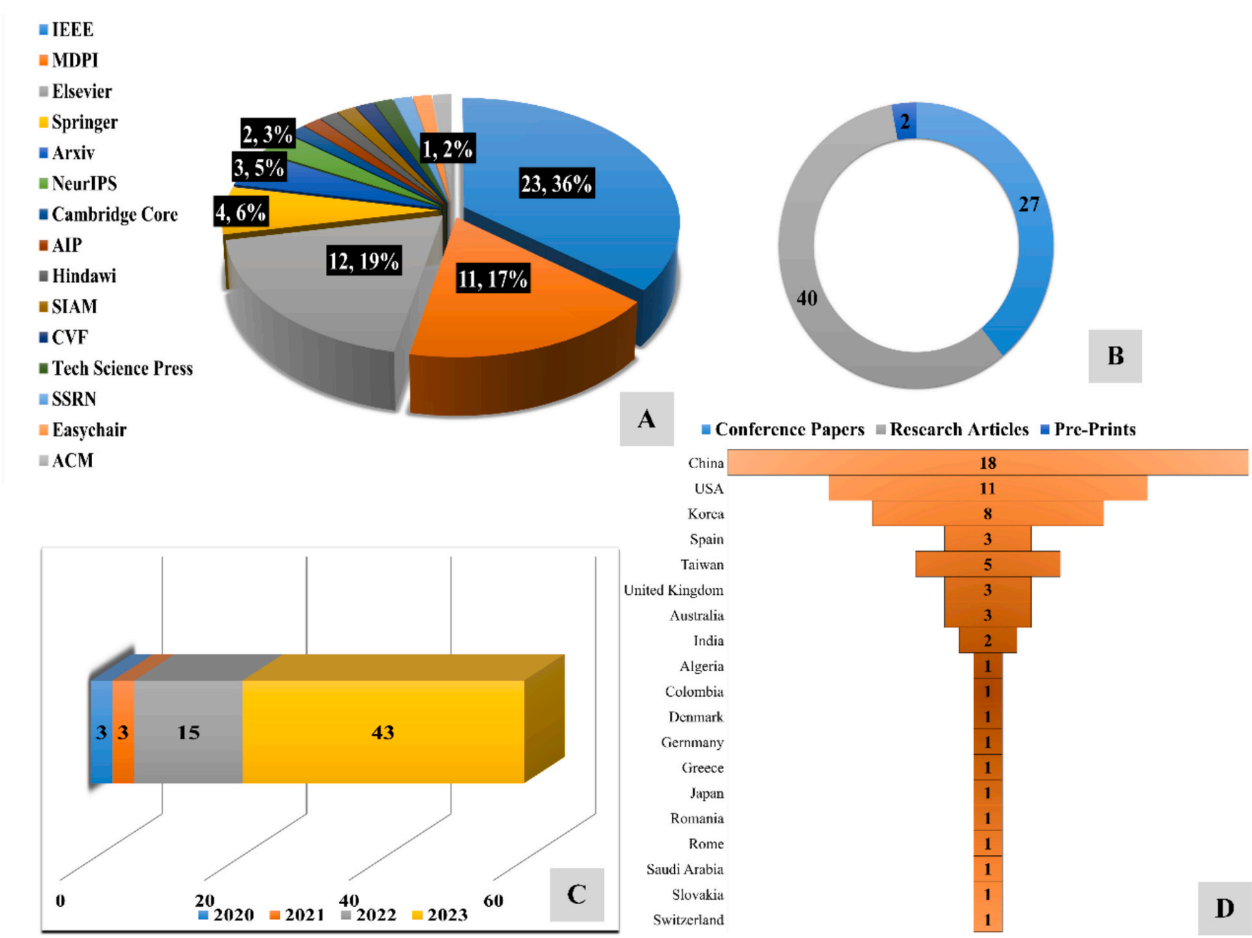

Fig. 1. Bibliometric analysis of Transformers research; (A) Publishers distribution, (B) Article Type distribution, (C) Yearly distribution and (D) Country-wise distribution.

telling. Peer-reviewed research articles, which make up 40 of the documented publications, underscore the rigorous investigative nature of the field, ensuring that the proposed forecasting methodologies withstand the scrutiny of the scientific method. This is complemented by the 27 conference papers stemming from influential platforms like NeurIPS and various IEEE-sponsored events, which highlight the collaborative spirit and the exchange of innovative ideas at the intersection of artificial intelligence and renewable energy. Pre-prints, though fewer in number, demonstrate the field's openness and progressive tilt towards early and broad dissemination of knowledge, with platforms such as Arxiv and Easychair facilitating this immediate exchange.

This uptrend in publication volume is particularly marked in the year 2023, where a significant surge to 43 publications from 15 in 2022 and just 3 in the preceding years manifests the rapidly growing interest in the application of this sophisticated computational technique to solar forecasting.

The geographical distribution of these scholarly endeavors paints a picture of a truly international effort. With China at the helm, contributing 18 publications, followed by the USA and Korea with 11 and 8 respectively, the research embodies a diverse yet focused array of global initiatives. Contributions from Spain, Taiwan, the United Kingdom, and Australia, with their respective outputs, further the narrative of a widespread, collective push towards optimizing solar energy forecasting. This global collaboration is enriched by single-paper contributions from countries across different continents, from Algeria to Switzerland, each adding to the kaleidoscope of research diversity. This broad participation underscores the universal relevance and urgency of advancing renewable energy forecasting methods.

Such a bibliometric review not only highlights the escalating pace of research activity in this critical area of study but also underscores the shared commitment across academia and industry to harnessing the potential of transformer models for a more sustainable future. The convergence of efforts from various publishers, conferences, and countries is a testament to the collective will within the research community to innovate and disseminate knowledge that can address one of the most pressing challenges of our time—energy sustainability.

2. Dichotomy of transformer models: Single and hybrid approaches in solar forecasting

In this section, the fundamental structure of transformer models is initially delineated, followed by an examination of their application in solar forecasting, categorized into both single and hybrid model approaches. These approaches are further segregated based on their distinct model structures. Transformer models have risen to prominence in solar forecasting owing to their adaptability and effectiveness. Within the single-model framework, the emphasis is on harnessing the intrinsic capabilities of the transformer for processing solar data. In contrast, the hybrid models integrate the transformer's self-attention mechanism with various computational methods, aiming to boost the accuracy of forecasts. Each approach, whether focusing on the pure application of transformer models or combining them with other techniques, provides valuable perspectives and benefits for predicting solar energy trends, underscoring the transformer's multifaceted utility in this niche area.

2.1. Basic transformer structure

The Transformer model represents a significant advancement in natural language processing, fundamentally changing the approach to machine understanding and language generation. Unlike its predecessors, such as Recurrent Neural Networks (RNNs) [57] and Long Short-Term Memory (LSTM) [58] models, the Transformer's unique architecture relies on self-attention mechanisms. The basic structure of Transformers consists of an encoder and decoder as depicted in Fig. 2 is as follows;

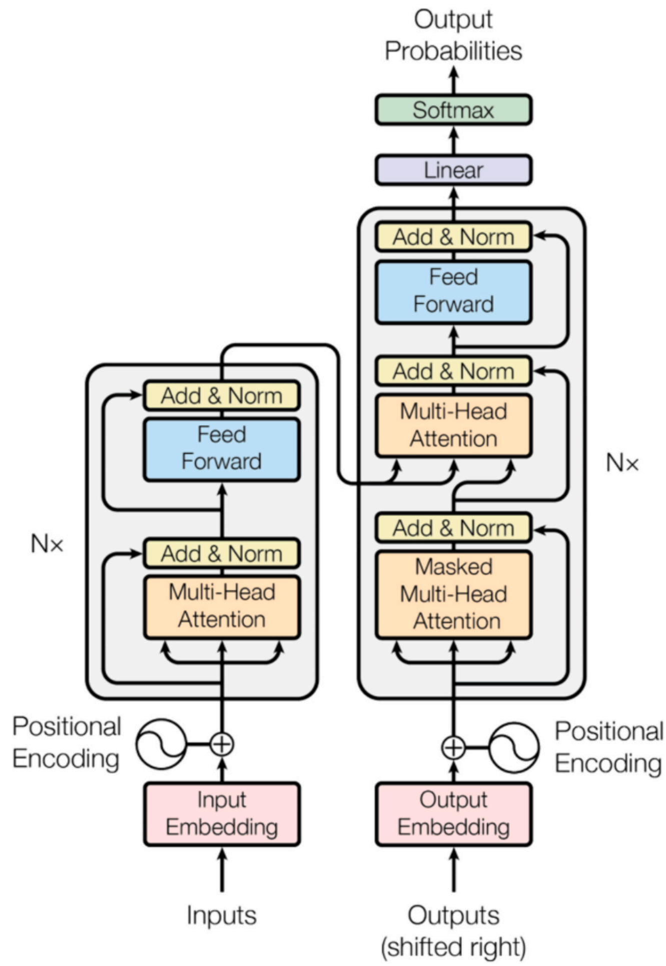

Fig. 2. Transformer - model architecture [41].

2.1.1. Multi-head self-attention mechanism

A pivotal element of the Transformer model, this mechanism variably weights different sections of the input sequence, which is essential for effective sequence processing. A thorough analysis of this mechanism is presented in a notable study titled "Analyzing the Structure of Attention in a Transformer Language Model" [59].

2.1.2. Positional encoding

To address the Transformer's inability to inherently recognize the order of sequences, unlike RNNs or LSTMs, positional encodings are incorporated into the input embeddings. This addition provides crucial information about the position of each word in the sequence.

$$\boldsymbol{PE}(\boldsymbol{pos}, \mathbf{2}\boldsymbol{i}) = sin\left(\frac{\boldsymbol{pos}}{\mathbf{10000}^{\frac{\mathbf{2}\boldsymbol{i}}{d_{model}}}}\right)$$

$$\boldsymbol{PE}(\boldsymbol{pos}, \mathbf{2}\boldsymbol{i}+\mathbf{1}) = cos\left(\frac{\boldsymbol{pos}}{\mathbf{10000}^{\frac{\mathbf{2}\boldsymbol{i}}{d_{model}}}}\right)$$

where $\boldsymbol{pos}$ is the position, $\boldsymbol{i}$ is the dimension and $\boldsymbol{d_{model}}$ is the embedding dimension of the model [41].

2.1.3. Encoder

The encoder in the Transformer model consists of two main sub-layers: a multi-head self-attention mechanism and a position-wise fully connected feed-forward network. This design facilitates the parallel processing of different parts of the input sequence, thereby significantly improving upon the efficiency of traditional RNNs [60]. The attention function computes attention scores using Query ***(Q)***, Key ***(K),*** and Value ***(V)*** vectors and the function is;

$$\boldsymbol{Attention}(\boldsymbol{Q},\boldsymbol{K},\boldsymbol{V}) = \boldsymbol{softmax}\left(\frac{\boldsymbol{Qk}^T}{\sqrt{\boldsymbol{d}_k}}\right)\boldsymbol{V}$$

Here, $\boldsymbol{d}_k$ is the dimension of keys [61].

Normalization and Residual Connections: Each sub-layer in the encoder is equipped with a residual connection and layer normalization. This configuration plays a key role in stabilizing the learning process [62].

2.1.4. Decoder

The decoder in the Transformer model is designed to mirror the encoder. It includes two sub-layers similar to those in the encoder and an additional sub-layer that performs multi-head attention on the output from the encoder stack. This feature allows the decoder to concentrate on relevant segments of the input sequence [63].

$$\boldsymbol{Multihead}(\boldsymbol{Q},\boldsymbol{K},\boldsymbol{V}) = \boldsymbol{Concat}(\boldsymbol{head}_i,\dots,\boldsymbol{head}_h)\boldsymbol{W}^O$$

$$\boldsymbol{head}_i = \boldsymbol{Attention}\left(\boldsymbol{QW}_i^Q,\boldsymbol{KW}_i^K,\boldsymbol{VW}_i^V\right)$$

where $\boldsymbol{W}$ matrices are the learnable parameters[64].

Just like in the encoder, each sub-layer of the decoder is connected through residual connections and is followed by layer normalization. This similarity ensures consistency in the processing approach across both components of the model [65].

2.1.5. Final linear and SoftMax layer

The output from the decoder is then passed through a final linear layer and a SoftMax layer. This stage is crucial for predicting the next word in the sequence, completing the model's language generation process [59].

2.2. Single model approaches

The utilization of single model approaches in solar forecasting, particularly focusing on transformer models, has gained significant traction. These approaches leverage the inherent strengths of individual transformer models, capitalizing on their specialized architecture to address the complexities of solar energy data. The cornerstone of these models is their encoder-decoder structure, which has been instrumental in effectively managing and interpreting the intricate layers of SI and related meteorological data.

2.2.1. Refinement of standard transformer structures

A hallmark of Transformer models is their adeptness at encoding diverse and extensive input data, followed by strategically decoding it to align with specific forecasting goals. This meticulous approach in data processing and analysis distinguishes transformer models, providing a detailed understanding of the fluctuating dynamics of solar energy.

Several studies exemplify this refinement and application of transformer models. Research on the Constrained Transformer highlights how traditional encoder-decoder layers are adept at systematically processing solar energy data [66]. Investigations into ultra-short-term PV power generation forecasting emphasize the importance of multi-headed attention in these models, enhancing accuracy by focusing on different data facets [67]. Comparative analyses pit transformer-based models against traditional time-series models like RNNs [57], underscoring the transformers' superior performance in solar radiation prediction [68]. These studies demonstrate the transformer's ability to handle a wide array of input parameters, showing its edge over conventional forecasting models. Further research delves into the adaptability of transformer models in real-time streaming contexts for short-term SI forecasting, showcasing their flexibility and rapid response capability in continuously changing data environments [69]. The integration of complex statistical analyses like the Pearson Correlation Coefficient (PCC) [70] for short-term PV generation forecasting further illustrates the versatility of transformer models, enhancing their predictive precision[71]. In-depth examinations, including comparative studies of deep neural networks for global horizontal irradiation forecasting, expand our understanding of how transformer models can be customized for specific forecasting tasks in the solar energy sector [72].

The synthesis of these diverse research efforts paints the standard transformer structure as a highly adaptable, efficient, and precise tool for solar forecasting. The consistent incorporation of encoder-decoder architectures [73], coupled with advanced features like multi-headed attention [74] and neural networks [75], emphasizes the suitability of transformer models for tackling the multifaceted challenges in solar energy forecasting.

2.2.2. Advancements in enhanced attention mechanisms

Further exploration of transformer structures in solar forecasting has led to innovations in refining the encoder-decoder architecture [76], optimizing attention mechanisms, and incorporating diverse data sets to improve forecasting methods. A central theme in these advancements is the foundational encoder-decoder architecture inherent to transformer models. For example, Attention-based transformer model utilizes a six-layered structure adept at handling long-distance dependencies, a notable improvement over LSTM models [77].Benítez et al. [78] and Wu et al. [79] further highlighted the transformer's efficiency in processing inputs in parallel, which reduces training time and bolsters the model's ability to capture long-term dependencies. This characteristic is crucial for managing temporal sequences in solar forecasting.

Attention mechanisms represent a vital aspect of these studies. Models by Zhou et al. [77] and Benítez et al. [78] incorporate multi-head attention mechanisms, allowing a nuanced focus on different facets of the input data, thus improving predictive accuracy. Self-attentive transformer architecture takes this further by optimizing the model specifically for SI data, showcasing an innovative approach to handling the variability inherent in solar energy inputs [80].

The inclusion of diverse data sets [81] and parameters also stands out as a key advancement. Jiang et al. [82] and Gao et al. [83] use of the Informer model, for instance, demonstrates its effectiveness in assimilating various meteorological parameters, like aerosol optical depth [84], to refine prediction accuracy. MATNet represents a hybrid approach, combining deep learning with physical-based models to achieve enhanced accuracy and robustness in forecasting[85]. These developments suggest that transformer models in solar forecasting are evolving towards increased complexity and sophistication. This evolution marks a shift from simple historical data processing to integrating a range of meteorological parameters and enhancing interpretability. Such progress indicates a trend towards models that are not only accurate but also versatile in adapting to the complex nature of solar energy data. However, this increasing complexity brings its own set of challenges. The computational demands of these sophisticated models, as pointed out by Yuan-Kang Wu [79], and the intricacies involved in their configuration, as noted by Pedro Lara-Benítez [78], highlight the need for a balance between complexity and practical feasibility. Additionally, the dependence on comprehensive and diverse data sets poses questions about the models' effectiveness in scenarios with limited data availability.

2.2.3. Innovations in adversarial training models

The advent of Adversarial Convolutional Transformer (ACT) [86] and Adversarial Sparse Transformer (AST) [87] models in time series forecasting marks a significant leap forward, particularly in the realm of solar energy predictions. Rooted in the standard transformer architecture, these models integrate novel elements that enhance their ability to navigate the complexities of multivariate time series data.

ACT[86] model incorporates several innovative modifications into the conventional transformer design as illustrated in Fig. 3. One pivotal

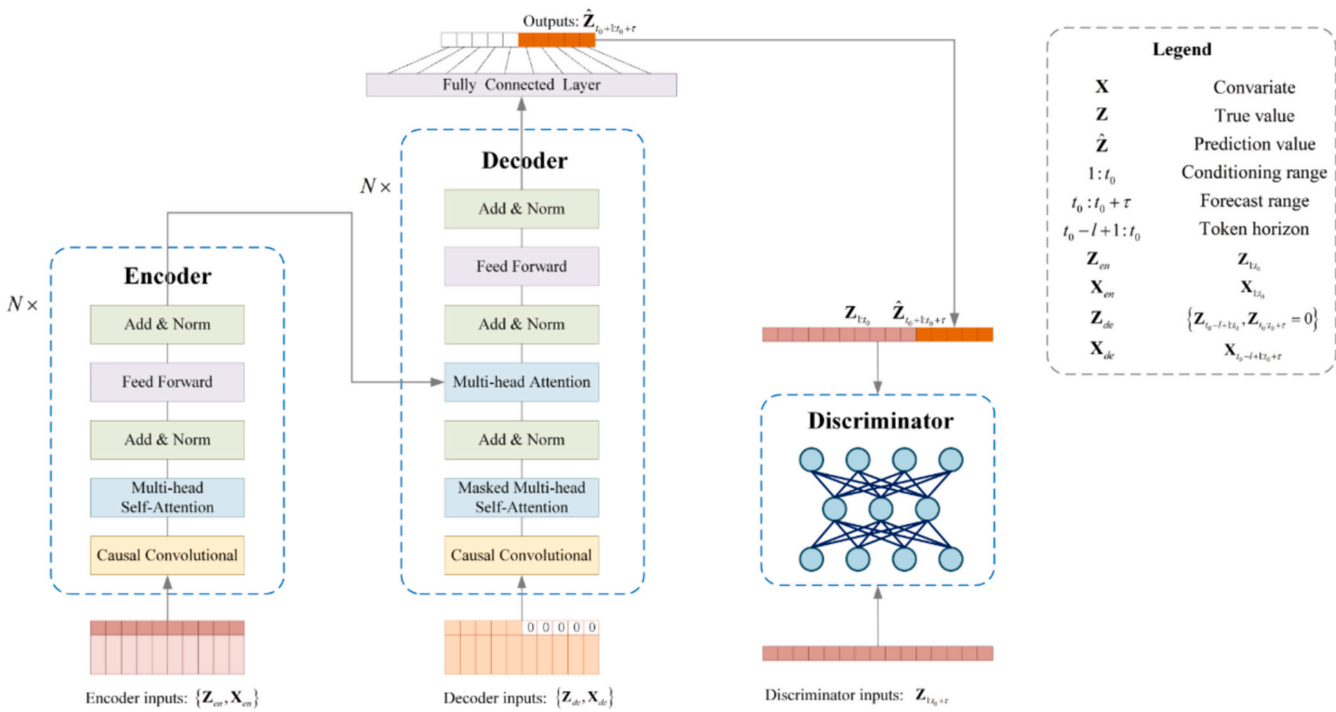

Fig. 3. Architecture of the Adversarial Sparse Transformer model [86].

change is the shift from stepwise decoding to a one-step method, which is instrumental in predicting entire sequences in a single forward pass, thereby significantly reducing error accumulation—a common issue in traditional models. Another notable enhancement is the convolutional attention block, which introduces local context into the self-attention mechanism, thus improving the model's proficiency in capturing short-term dependencies. Furthermore, the model employs adversarial training[88], a technique designed to recognize long-term repeating patterns, a crucial aspect for accurate solar forecasting where cyclical patterns are influenced by diverse environmental factors.

AST [87] model brings a unique dimension to transformer architectures by substituting the standard softmax [89] function in the attention heads with α-entmax [90]. This alteration results in sparse attention weights, which are more effective in concentrating on pertinent parts of the input data. The selection of $\alpha = 1.5$ strikes a balance between sparsity and efficiency. Drawing inspiration from Generative Adversarial Networks (GANs) [91], the sparse Transformer [92] concept introduces sequence-level regularization [93], a key attribute for long-horizon forecasting. This feature is particularly beneficial in solar energy forecasting, where long-term predictions with high accuracy are invaluable. Both the ACT and AST models have demonstrated superior performance in various experiments, outshining traditional transformers and other baseline models in solar energy forecasting tasks. The ACT model, with its one-step decoding, convolutional attention block, and adversarial training, excels in capturing both short-term and long-term patterns in solar data. Concurrently, the AST model's sparse attention mechanism introduces a heightened level of precision and focus, significantly boosting its forecasting accuracy.

These innovative models represent a pivotal shift in the development of transformer models for time series forecasting. The integration of adversarial components and sparse attention mechanisms indicates a progression towards more intricate and refined models. These models are adept at handling the complexities of solar energy data and exhibit enhanced capabilities in forecasting across various time horizons with greater accuracy. Such advancements are crucial in the renewable energy sector, where precision and dependability in predictive technologies are of utmost importance.

2.2.4. *Encoder-decoder variations*

The field of solar forecasting is witnessing a significant shift with the introduction of advanced transformer models that employ innovative architectures, aiming to increase the accuracy and comprehensiveness of predictions. These developments, including the Dual-Encoder Transformer (DualET) [94], Temporal Fusion Transformer (TFT) [51], and Vision Transformers (ViTs) [95], represent more than just incremental advancements; they are pivotal innovations in solar energy forecasting.

A notable example is DualET model [94] depicted in Fig. 4, which introduces a dual-encoder setup comprising a Local Seasonal Information (LSI) encoder and a Remote-Sensing Information (RSI) encoder, paired with a single decoder. This distinctive configuration, capable of processing both sequential and image data, marks a significant evolution from traditional transformer models. The LSI encoder captures temporal dynamics from ground-based measurements, while the RSI encoder processes spatial-temporal features from satellite imagery. This comprehensive approach allows for an in-depth analysis of solar power fluctuations, with a cross-domain attention module enhancing the model's ability to correlate sequence data with cloud imagery. The DualET model's effectiveness in short-term PV power prediction highlights its strength in integrating various data types for more refined forecasting.

Similarly, TFT [51] is designed for multi-horizon forecasting [96], skillfully balancing known and future variables, reflecting the transformers' evolution in managing the uncertainties and variabilities in solar data. Application of large pre-trained ViTs [95] for solar irradiation prediction is another stride forward. Using transfer learning [97] from an extensively trained model underscores the importance of cross-domain applicability and the adaptability of transformer models to different environmental and geographical contexts. The model of iTransformer [98], which innovatively inverts the roles of the attention mechanism[99] and feed-forward network, demonstrates the continuous refinement of transformer components to enhance solar forecasting performance.

These advancements collectively signify a trend towards transformer models that are not only more accurate but also proficient in handling the diverse and complex nature of solar energy data. The integration of

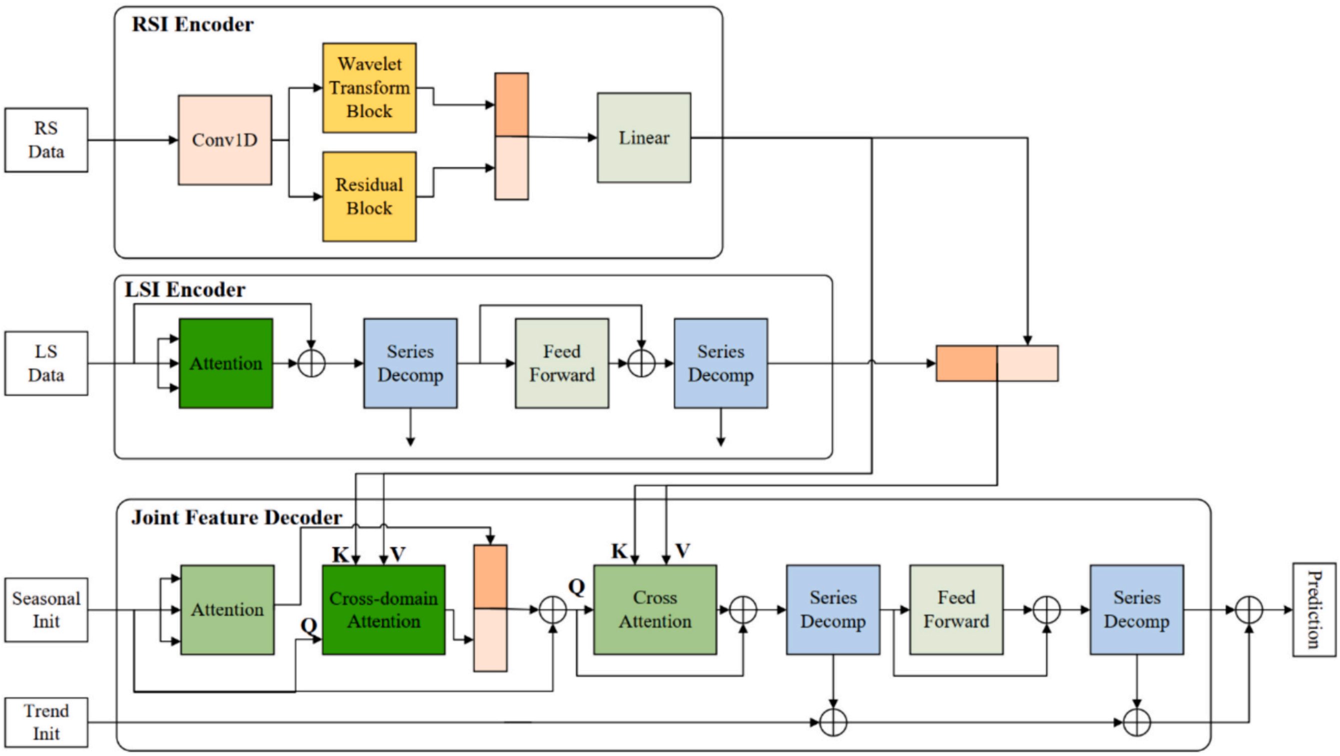

Fig. 4. Model architecture of DualET [94].

image and sequence data processing, the focus on multi-horizon forecasting, and the utilization of transfer learning are crucial steps in evolving transformer models. These models are increasingly equipped to effectively tackle the variability and uncertainty in SI and power generation, paving the way for more reliable and efficient solar energy forecasting.

2.2.5. *Encoder-only models*

A significant evolution in transformer model application to SI forecasting is evident with the development of the Multi-Branch ResNet-Transformer (ResTrans) models [100,101]. Originally utilized in natural language processing (NLP) [102], these models have been adeptly repurposed to address the unique challenges of solar energy prediction. This innovative adaptation involves a departure from the standard transformer structure. The ResTrans models, moving away from the conventional encoder-decoder format typical in sequence-to-sequence learning [103], employ a modified architecture tailored for processing sequential solar data and generating single predictive outputs. This strategic shift is crucial, aligning the models' capabilities more closely with the nuanced requirements of solar forecasting tasks.

A notable structural modification in the ResTrans models shown in Fig. 5 is the replacement of NLP-centric word embedding with time embedding. This change significantly enhances the models' proficiency in capturing temporal patterns in SI data. Additionally, the inclusion of a 1-dimensional convolutional neural network (1D CNN) [104] layer, substituting the standard feed-forward layer, allows these models to more effectively discern local variations and patterns in the solar data.

The unique multi-branch design of these models involves individual processing of data from separate solar sites, culminating in an aggregated analysis via a ResNet model. This novel approach adeptly addresses both spatial and temporal dependencies in SI data, ensuring a comprehensive and localized understanding of solar energy dynamics across various geographical regions. This innovative architecture underscores the potential of encoder-only models in enhancing the accuracy and relevance of solar forecasting, reflecting a significant advancement in the application of transformer technologies to the field of renewable energy.

2.2.6. *Probabilistic and generative models*

The evolution of transformer models in SI forecasting is further exemplified by the work of researchers like Tong et al. [105,106], Jawed et al. [107], Jønler et al. [108], Sharda et al. [109], Mercier et al. [110], and Sinha et al. [111,112], who have steered these models towards more complex and nuanced architectures. These advancements specifically cater to the unique challenges of solar energy prediction.

A notable advancement is the integration of hierarchical and

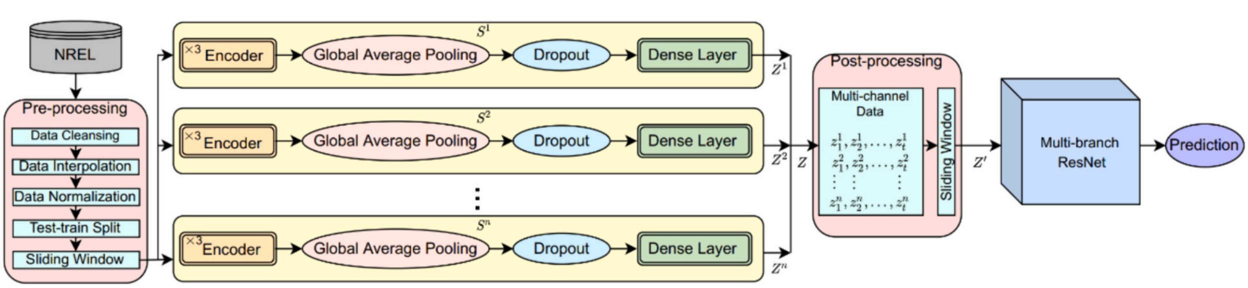

Fig. 5. Spatio-temporal multi-branch ResNet-Transformer architecture [100].

probabilistic forecasting methods within the transformer framework. Probabilistic Decomposition Transformer (PDTrans) [105,106] and Probabilistic SI Transformer ProSIT [108] model are prime examples. PDTrans [105,106] combines the transformer structure with conditional generative models and leverages variational inference for hierarchical forecasting, adeptly managing complex, multi-tiered data structures. ProSIT [108] integrates a probabilistic module into its transformer architecture, focusing on multi-horizon forecasting and introducing a gated residual layer to effectively handle zero-value scenarios.

GQFormer [107] model innovatively incorporates a multi-quantile generative approach [113] within the transformer architecture. This model produces multiple quantile estimates for each forecasting horizon by embedding quantile levels into the transformer's self-attention mechanism, broadening its range of potential outcomes. RSAM [109] model uses transformers' self-attention mechanism for multi-horizon forecasting, integrating quantile regression for prediction interval modeling. This approach enhances the models' interpretability and robustness by not only predicting values but also providing a range of probable outcomes. SI Anticipative Transformer (SIAT) [110] represents a significant departure from traditional transformer models. Employing a self-attention-based network with a GPT-2 [114] based decoder, it adeptly processes image data, showcasing the transformer's versatility in handling different data types. Sinha et al.'s work [111,112] in extending the forecasting horizon of transformer models highlights their scalability and flexibility, maintaining structural integrity across various forecasting scenarios.

These developments collectively signal a transformation in transformer architecture, moving from a one-size-fits-all approach to more specialized, context-sensitive designs. The integration of hierarchical and probabilistic methods, multi-quantile forecasting, image data handling, and extended forecasting capabilities reflect a deeper comprehension of the intricacies inherent in SI data. These structural innovations indicate a broader trend of developing transformer models as comprehensive tools, well-equipped to analyze and interpret the complex dynamics of solar energy.

2.2.7. Specialized models for distinct domains or applications

Solar energy forecasting is witnessing transformative advancements with the development of specialized transformer models tailored for specific applications. A notable example is Conformer-GLaplace-SDAR model [115], which innovatively combines the Conformer [116] model with statistical distribution analysis for solar radiation prediction. Utilizing fast Fourier transform for multi-variable correlation extraction, this model significantly advances transformer architectures by efficiently handling multi-scale dynamics and temporal patterns at various scales, focusing on computational efficiency and stability. The integration of the SDAR model for interval prediction across diverse confidence levels further enhances the Conformer's forecasting capabilities, making it especially relevant for SI forecasting. Similarly, Gao's et al. [117] transformer-based model, integrating a clear sky model [118] for short-term SI prediction, demonstrates an added layer of domain-specific refinement. By focusing on residual prediction between actual and clear sky model irradiance values, this approach marks a shift towards models that integrate contextual environmental variables for more precise forecasting.

Further illustrating the versatility of transformer architectures, SolarFormer [119] and VIT [120] model showcase their adaptability in processing a variety of data types, including imagery and spatial sensor data. SolarFormer employs a multi-scale Transformer encoder and a masked-attention Transformer [121] decoder, an innovative approach for processing solar PV data [119]. In contrast, Kang et al. [120] VIT model leverages multi-head self-attention for auxiliary data processing from surrounding PV sensors, demonstrating an acute sensitivity to the spatial aspects of solar data.

These developments collectively underscore the growing trend of customizing transformer structures to meet the specific demands of solar energy forecasting. This customization involves the integration of multi-scale dynamics, application of domain-specific knowledge such as clear sky models, and modification of the architecture to accommodate different data types like spatial sensor data and imagery. These enhancements go beyond mere incremental improvements, representing a significant paradigm shift towards the creation of comprehensive, context-aware models. These models are designed to capture the intricate dynamics of SI and PV energy production, reflecting a deep understanding of the multifaceted nature of solar energy forecasting.

2.2.8. Models with novel data handling or integration techniques

In the evolving landscape of renewable energy forecasting, Graph Patch Informer (GPI) [122] model emerges as a groundbreaking innovation. It ingeniously blends elements from the Informer Encoder [123] with self-supervised learning during its pre-training phase as shown in Fig. 6. This unique approach equips the model to grasp abstract-level representations, a critical factor for decoding complex temporal patterns prevalent in renewable energy data. The GPI model further innovates by incorporating a Graph Attention Network (GAT) [124] with a self-adaptive adjacency matrix during its fine-tuning stage, enhancing the flow and assimilation of information across temporal representations. The integration of GAT within the GPI model is a significant architectural advancement, allowing for a deeper and more nuanced understanding of interrelated data patterns.

SL-Transformer [125] model represents another stride forward in this field. This model synergizes the Savitzky–Golay filter [126] and Local Outlier filter [127] with LSTM and Transformer structures. Its aim is to fully harness multimodal features, such as wind speed, historical power data, and solar power data, for enhanced energy forecasting. Integrating these filters into the model's architecture is pivotal for refining data inputs, addressing the common challenges of noise and outliers that often compromise the accuracy of renewable energy forecasting. Additionally, the SL-Transformer's incorporation of a self-attention mechanism boosts its capability to dynamically modify input data weights, marking a significant shift towards more adaptable and precise forecasting models.

The GPI [122] and SL-Transformer [125] models exemplify a wider movement in renewable energy forecasting, characterized by the integration of domain-specific techniques and customizations into transformer architectures. These innovations represent a shift from merely applying the transformer framework to new datasets, to a more profound re-envisioning of the transformer structure, tailored to meet the specific requirements of renewable energy forecasting. This includes augmenting the model's ability to handle abstract representations, process multimodal data effectively, and adapt to the unique temporal dynamics of solar and wind energy.

The variations on single transformer models in solar forecasting reveal a dynamic progression, showcasing significant innovations and customizations. This includes the refinement of standard transformer structures with advanced attention mechanisms and the emergence of encoder-only models. Innovations extend to probabilistic and generative models, which bring a nuanced approach to solar energy forecasting. Specialized models are tailored for distinct domains or applications, demonstrating a keen understanding of the specific challenges in solar forecasting. Additionally, models featuring novel data handling and integration techniques mark a significant shift, integrating elements like GAT and multi-scale dynamics. These developments represent a move towards transformer models that are not only more sophisticated and precise but also uniquely adapted to the complex nature of solar energy data, highlighting their versatility and growing effectiveness in the renewable energy sector.

2.3. Hybrid model approaches

In solar forecasting, there is a notable shift towards developing custom hybrid models that prominently incorporate transformers,

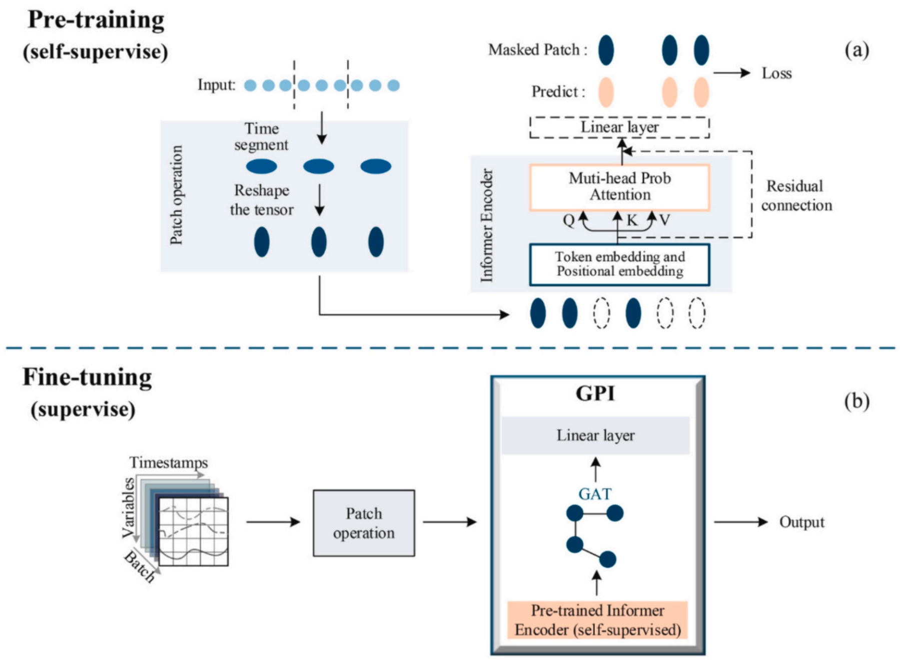

Fig. 6. The overall architecture of the proposed GPI. (a) Pre-training performing self-supervised learning. (b) Fine-tuning performing supervise learning [122].

reflecting the multifaceted challenges of this field. These hybrid models ingeniously merge transformer technology with various advanced techniques, such as convolutional neural networks (CNNs)[128] and LSTM networks[129]. The ability of transformers to process long sequences and handle complex dependencies makes them a core component of these models. They are synergized with other computational strategies to boost prediction accuracy and adaptability, signifying a major evolution in renewable energy forecasting. This integration marks a departure from traditional methods, moving towards more intricate and adaptable models capable of navigating the complexities of solar data. The focus on transformers in these hybrid models underscores their vital role in enhancing solar energy forecasting, leading to more nuanced and comprehensive analytical tools in renewable energy.

2.3.1. Transformer and CNN hybrid models

One such innovative approach is seen development of the Convolutional Transformer model [130]. This model combines 1D dilated causal CNNs [131] with transformer encoder and decoder blocks, enhancing temporal feature extraction and adaptability to data with periodic zero and non-zero values. This combination offers a dual advantage: CNNs capture local temporal dependencies effectively, while transformers manage global dependencies, crucial for the nuanced nature of solar energy data. Hybrid CNN-LSTM-Transformer [132] model further extends this methodology. By integrating LSTM networks, the model aims to capture long-term dependencies in time series data more effectively. Season-based clustering techniques for preprocessing and tailored modifications in the transformer's self-attention block are designed to accommodate seasonal variations in solar irradiation.

CT-NET [133] model presents a unique blend of CNNs and transformers, using a CNN-based multi-head attention mechanism. This model is focused on extracting both local and contextual features, enhancing the accuracy and efficiency of solar power forecasting. It reflects an understanding that weather-related data demands a combination of local feature extraction (via CNNs) and global context comprehension (via transformers). SpringNet [134] introduces an innovative Spring DTW attention mechanism [135] within the transformer framework, concentrating on the shapes of time series patterns, essential for accurate solar power generation forecasting. The model employs a GPU-accelerated Spring DTW algorithm, addressing the computational intensity often associated with transformers, thereby enhancing their practicality for real-world applications.

2.3.2. Transformer and RNN hybrid models

Combining Transformer architectures with RNN methods is also a growing trend in solar forecasting. This blend aims to harness the sequential data processing capabilities of RNNs while integrating the sophisticated attention mechanisms of Transformers. This approach is particularly effective in addressing the complex and variable nature of solar energy data.

ISt-LSTM-Informer [136] model stands as a prominent example of this trend. It combines LSTM (a type of RNN) with the Informer (a Transformer-based model) within a stacking ensemble framework. This hybrid model is adept at capturing both short and long-term temporal dependencies in PV power forecasting. It leverages the LSTM's strength in handling short sequence time-series data and the Informer's proficiency in managing longer sequences.

DTTrans [137] model represents another innovative approach as illustrated in Fig. 7, merging Delaunay triangulation [138] with a Transformer-GRU (Gated Recurrent Unit) [139] hybrid. This model selects weather data from stations forming a Delaunay triangle around a target PV site and employs a TransGRU model. This hybridization synergizes the Transformer's attention mechanisms with the GRU's sequential data processing abilities. The DTTrans model aims to provide robust forecasting against weather forecast errors, ensuring interpretability and adaptability to various PV sites.

Phan et al.'s [140] approach is another illustration of the hybrid model concept, combining a modified Transformer with extreme Gradient Boosting (XGBoost) for error correction. This model utilizes XGBoost [141,142], a ML algorithm, to address missing data issues and employs the Transformer for prediction. This approach represents a marriage of traditional ML techniques with advanced deep learning models. TEFNEN [143] model integrates the functionalities of traditional RNNs with the advanced attention mechanisms of Transformers. TEFNEN adapts the Transformer architecture for energy forecasting by incorporating RNN layers in place of the Transformer's original positional encoding. This modification aims to enhance the model's ability to process historical values in time-series data, exemplifying the innovative fusion of different neural network methodologies in solar energy

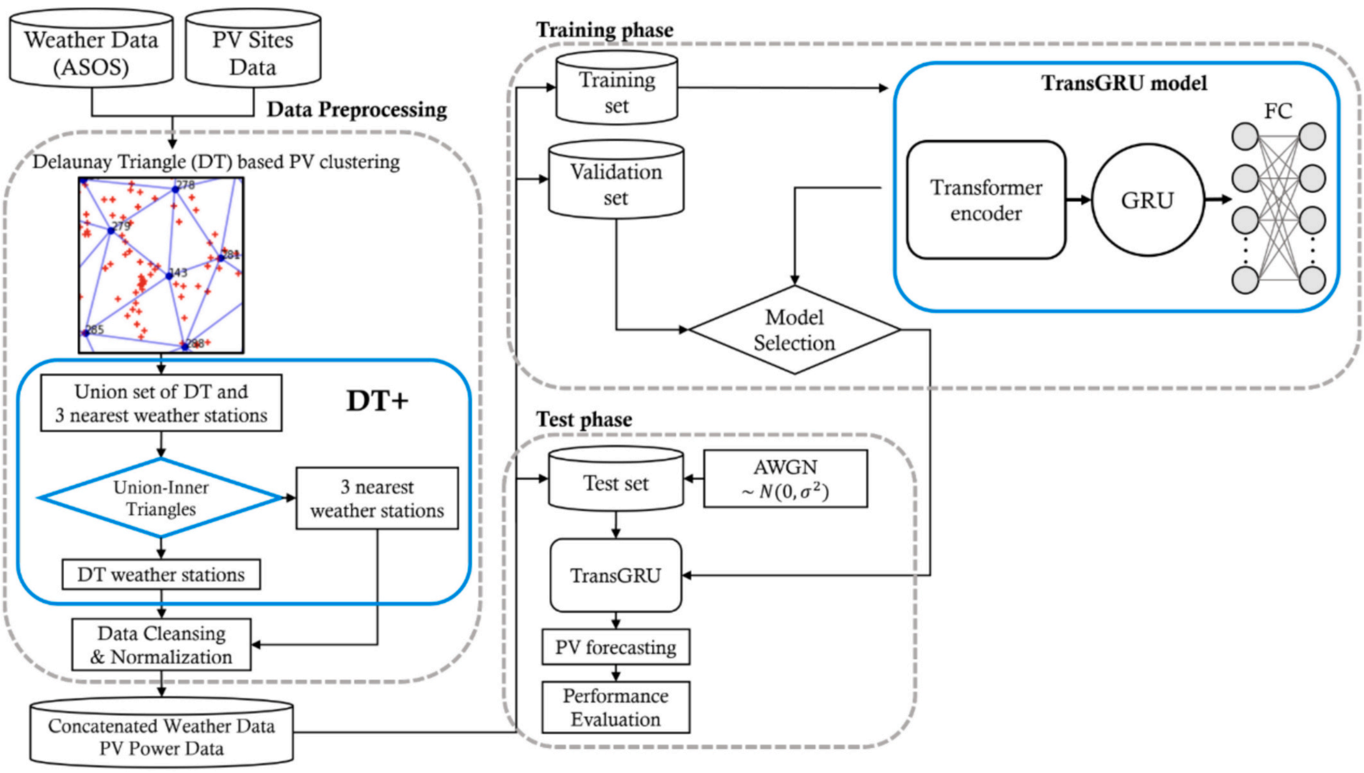

Fig. 7. Overall process of the DTTrans framework [137].

forecasting.

2.3.3. Transformer and statistical/physical models hybrid

The recent research by Padilha et al [144], Gerges et al. [145,146], Mouloud et al. [147], and Santos et al. [148] marks a significant trend in renewable energy forecasting, where hybrid models that combine classical time series forecasting techniques with Transformer-based deep learning architectures are developed. These innovative models aim to harness the strengths of both traditional and modern approaches to tackle the complex challenges of renewable energy prediction.

H-Transformer model [144] is a prime example of such a hybrid approach. It uniquely merges the traditional SARIMA [149] model, known for its linear forecasting capabilities, with a Transformer that focuses on nonlinear residual series forecasting. The integration of Bayesian Hyperparameter Optimization [150] in this model further refines its accuracy, enabling it to capture both linear trends and complex nonlinear patterns in renewable energy forecasting. DeepSI model, presented in two distinct studies [145,146], adopts a pioneering hybrid approach. It combines bidirectional long short-term memory (BLSTM) [129,151] autoencoders with Transformers and enhances the model's performance with uncertainty quantification through Monte Carlo dropout [152] sampling. This architecture is adept at capturing temporal dependencies and intricate relationships within climate simulation data, a critical component for accurate long-term SI forecasting.

CS-TFT model [147] incorporates the McClear sky model [118] into a Transformer-based framework. Employing the TFT approach, this model excels in forecasting Global Horizontal Irradiance (GHI) across different time steps and climatic zones. The CS-TFT model's consideration of both historical GHI data and clear sky conditions exemplifies its adaptability and precision in forecasting. Santos's research utilizes the TFT model [148] for day-ahead PV power forecasting, demonstrating the effectiveness of merging deep learning with temporal fusion techniques. The TFT model employs specific components for attribute selection and gating layers, which help in filtering out non-essential elements, thereby ensuring a high-performance output in PV power forecasting. These developments collectively highlight a shift towards more sophisticated, hybrid forecasting models that integrate the best of both statistical and deep learning techniques, catering to the nuanced requirements of renewable energy forecasting.

2.3.4. Transformer and graph neural network (GNN) hybrid models

Simeunović et al. [153] research introduces an innovative approach to PV power forecasting by leveraging spatio-temporal Graph Neural Networks (GNNs)[154]. This strategy is particularly focused on multi-site PV power forecasting, adeptly addressing the intricate spatial and temporal factors influencing PV energy production. Central to this research are two novel models: the Graph-Convolutional Long Short-Term Memory (GCLSTM) [155] and the Graph-Convolutional Transformer (GCTrafo) [153]. These models conceptualize PV production time-series as graph signals, using graph-based models to harness the spatio-temporal dependencies within PV production data. This approach is exceptionally beneficial for densely networked PV systems, enabling the models to decipher cloud dynamics and enhance forecasting accuracy.

GCLSTM and GCTrafo diverge in their handling of time dependence. GCLSTM employs recurrent structures, a staple in time-series modeling, particularly adept at capturing sequential data patterns. GCTrafo, on the other hand, utilizes attention mechanisms, an innovation in machine learning known for its efficacy in Transformer-based models (Fig. 8). This mechanism allows GCTrafo to focus selectively on various spatial or temporal aspects of the data, proving advantageous for longer prediction horizons. GCTrafo, drawing inspiration from the base Transformer architecture, incorporates specific adaptations for the multi-site PV generation forecasting problem. In contrast to GCLSTM, it eschews recurrent structures in favor of a multi-head attention approach, where each head targets different patterns in the input sequence. This feature is pivotal in managing the complex spatio-temporal data encountered in PV forecasting.

While both models employ graph convolutional layers to decipher spatial patterns, they differ in their temporal processing approaches.

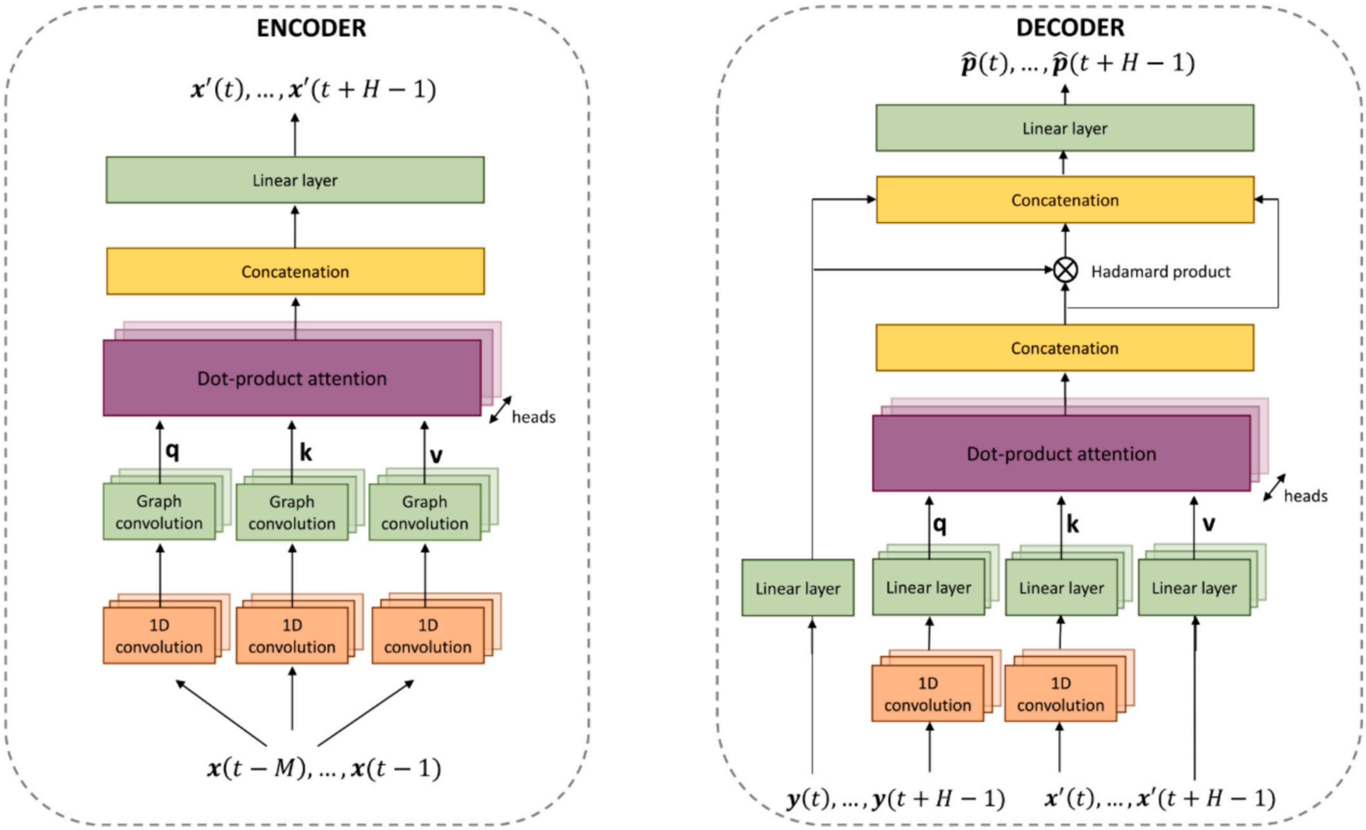

Fig. 8. Graph Convolutional Transformer architecture [153].

GCLSTM might be more suitable for shorter prediction horizons due to its recurrent structures, whereas GCTrafo, with its attention mechanism, is potentially more effective for longer-term forecasts.

The results from these models, especially GCTrafo, indicate strong performance across various forecasting horizons, underscoring the efficacy of GNNs in capturing spatio-temporal correlations in PV production data. The attention-based approach of GCTrafo allows itself to focus more efficiently on different spatial or temporal information than recurrent-structure models.

Simeunović et al. [153] underscored the potential of graph convolutional approaches in renewable energy forecasting. By marrying the spatial insights provided by GNNs[154] with the temporal acumen of LSTM and Transformer mechanisms, these models mark a significant leap in multi-site PV power forecasting. This hybrid methodology, combining graph-based modeling with advanced time-series analysis techniques, holds promise for enhancing the precision and efficiency of PV power forecasting, particularly in grid management scenarios involving multiple nodes.

2.3.5. *Custom hybrid models with multiple techniques*

Previous studies highlighted a trend towards integrating diverse methodologies within a single predictive framework for PV power forecasting. The relevant hybrid models are characterized by their innovative blend of techniques, each contributing unique strengths to address the multifaceted challenges of PV power forecasting.

The model of VMD-ECGO-LSH-Informer [156] presents a sophisticated approach to PV power forecasting. This hybrid model integrates several advanced techniques to refine prediction accuracy and efficiency for PV power signals illustrated in Fig. 9. The core of this model is the Variational Mode Decomposition (VMD) method [157,158], which simplifies the PV power sequence by breaking it down into Intrinsic Mode Functions (IMFs) [159] and a residual term. This decomposition helps in reducing the volatility and non-stationarity of the PV power signal, making it easier to analyze and predict. Enhancements to the Informer model, a deep learning architecture known for its effectiveness in handling long-range dependencies, are achieved through Locality Sensitive Hashing (LSH)-based [160] attention. This innovative attention mechanism employs random sampling to identify key vectors relevant to query vectors, thereby improving the model's accuracy. Further sophistication is added through the Enhanced Chaos Game Optimization (ECGO) algorithm [161]. It introduces novel elements like circle chaos initialization and the golden sine algorithm for the optimization process. This enhances the robustness and efficiency of the model, particularly in tuning its hyperparameters.

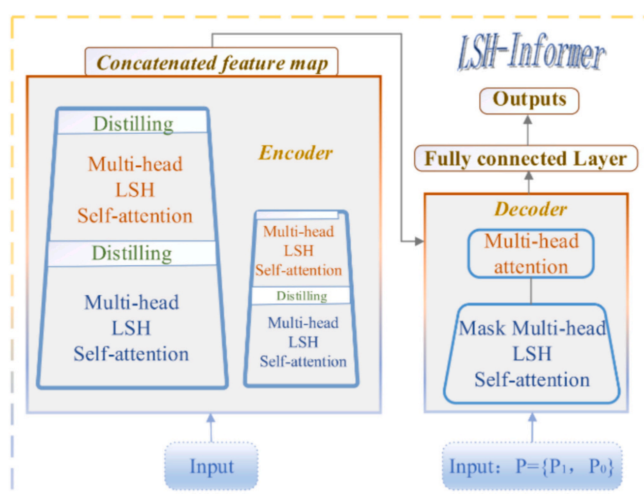

Fig. 9. Framework of VMD-ECGO-LSH-Informer [156].

The model of Transformer-LUBE [162] represents a novel blend of Transformer architecture with Lower Upper Bound Estimation (LUBE) [163] for probabilistic forecasting. This model combines deterministic forecasting through the Transformer with interval forecasting via LUBE, further enhanced by pre-processing and post-processing techniques like missing data imputation using XGBoost and error adjustment with a GRU. The model of QT-MARF [164] integrates Quantile Transformation with a Multi-Attention Transformer Framework. This innovative combination ensures a more uniform distribution of input data, facilitating learning, while the addition of residual networks and gated residual networks improves the model's capacity to discern complex patterns.

The model of VIT-GRU [165] merges the ViT with the GRU, combining spatial and temporal processing capabilities. This synergy allows for an in-depth analysis of spatial features from sky images and temporal relationships in time series data, making it a robust forecasting tool. Lastly, the model of Liu et al. [166] employs a dynamic feature encoder based on ViT, complemented by cross-modality attention. This approach exemplifies the emerging trend in multimodal learning frameworks for PV forecasting, emphasizing the importance of integrating diverse data types, such as sky images and GHI series, for enhanced forecast accuracy.

The development of custom hybrid models that incorporate a diverse range of techniques, addressing the multifaceted challenges of PV power forecasting. These models, exemplified by Peng et al. [156], Phan et al. [162], Mirza et al. [164], Xu et al. [165] and Liu et al. [166], blend advanced methods like the Variational Mode Decomposition, Enhanced Chaos Game Optimization, Locality Sensitive Hashing, and various deep learning architectures. The models aim to refine prediction accuracy and efficiency, handling complex temporal and spatial data characteristics inherent in solar energy. Key innovations include combining transformers with probabilistic forecasting methods, integrating spatial and temporal data processing, and employing multimodal learning frameworks. These developments represent a significant trend towards creating more sophisticated, adaptable, and precise forecasting tools in solar energy, leveraging the strengths of multiple computational strategies to enhance the accuracy and reliability of PV power forecasting.

3. Temporal dynamics in solar forecasting: Dissecting time frames and output variables

This section provides a comprehensive analysis, presented visually as a figure, that captures the deployment of transformer models across various forecasting timeframes in solar energy research. The data, categorized by the forecasting period and output parameters, offers a snapshot of current research trends.

Fig. 10 reveals a predominant focus on short-term forecasting, with 47 articles exploring transformer model applications in this category. This is followed by very short-term forecasting, which is the subject of 12 articles, underscoring the growing interest in immediate solar forecasting needs. Medium-term forecasting and unspecified timeframes are less represented, with 2 and 3 articles, respectively, indicating potential areas for future research.

In terms of output parameters, the majority of the research, amounting to 29 articles, is dedicated to PV power forecasting, demonstrating a significant interest in this area. This is closely followed by 25 articles concentrating on GHI or SI forecasting, highlighting the importance of these parameters in the field. The research corpus also includes 9 articles that focus on time series data derived from solar datasets, illustrating the diversity of data handling in transformer models.

Additionally, the discussion encompasses the statistical metrics like MAE (Mean Absolute Error), MSE (Mean Squared Error), RMSE (Root Mean Squared Error)[167], nMAPE (Normalized Mean Absolute Percentage Error), R2 (Coefficient of Determination)[168], PCC (Percentage of Consonants Correct), CRPS (Continuous Ranked Probability Score)[169], NRMSE (Normalized Root Mean Squared Error) and NMAE (Normalized Mean Absolute Error)[168] are used to evaluate the performance of these models, offering a comprehensive view of their effectiveness and accuracy. These metrics are vital in assessing the models' proficiency and comparing them with traditional or other advanced forecasting models. The holistic detail is available as Appendix-1, a snippet of the Appendix is presented here as Table 1 to provide a clear picture. This data visualization and analysis underscore the significant traction transformer models are gaining in solar forecasting, mapping out the landscape of current research efforts by time frame, output parameter, and modeling approach. This overview not

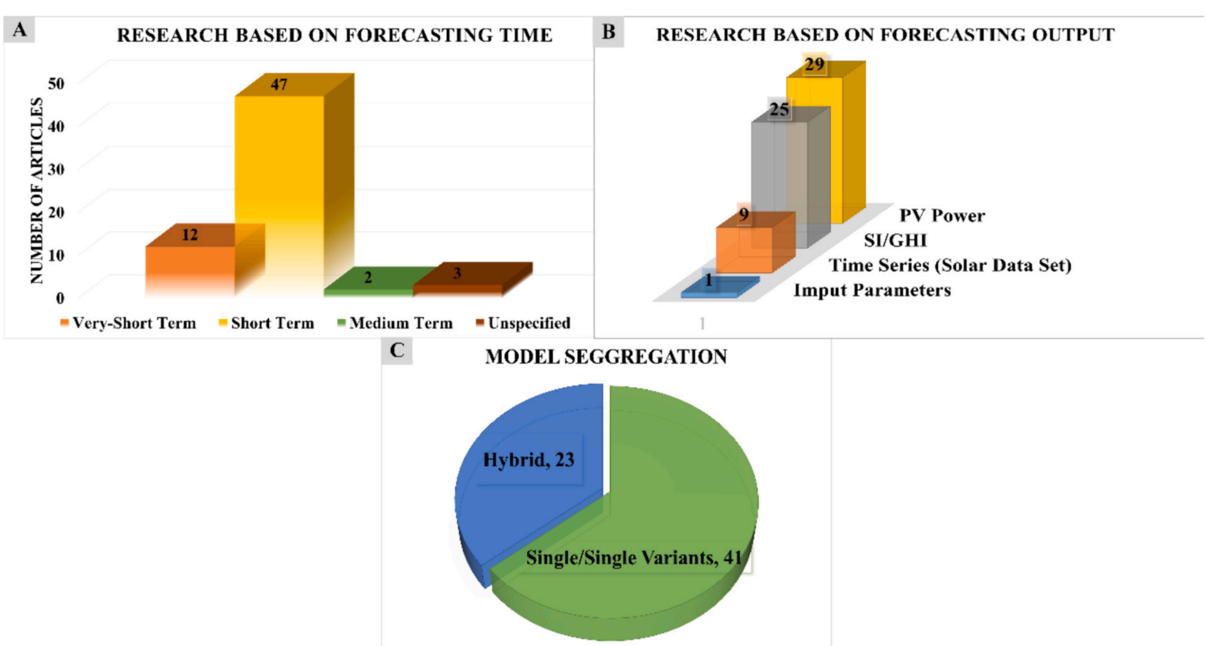

Fig. 10. (A) Research based on forecasting time, (B) Research based on forecasting output and (C) Model segregation.

Table 1
Models' parametric details, comparison and validation.

Model Name	Model Type	Prediction	Parameters	Comparison	Forecasting Time	Statistical Validation
RSAM (Robust Self-Attention Multi-horizon) [109]	Single	GHI	Dew point Cloud type Wind speed Wind direction Relative humidity Precipitable water Temperature Pressure Solar zenith angle	Smart Persistence Algorithm LSTM A-LSTM CNN-LSTM A-CNN-LSTM	1-h 2-h 12-h	**India Data Set** RMSE: 52.45 MAE: 26.23 FS: 54.57 **Hotevilla Data** RMSE: 64.87 MAE: 30.32 FS: 61.2
SpringNet [134]	Hybrid	Multivariate Time Series	Solar power generation data Temperature Humidity DHI Month hour-of-the-day minute-of-the-hour	DeepAR LogSparse Transformer Persistence model	Hourly	**Sanyo** MAE: 0.258 RMSE: 0.380 **Hanergy** MAE: 0.358 RMSE: 0.494
H-Transformer (SARIMA -Transformer) [144]	Hybrid	PV Power	Weather data Energy generated	SARIMA RNN LSTM GRU Transformer SARIMA-RNN SARIMA-LSTM SARIMA-GRU	Hourly	**Solar-1** RMSE: 766 MAE: 516 **Solar-2** RMSE: 1307 MAE: 934
Conformer [115]	Single	SI	Daily Sunshine Duration Average Relative Humidity Daily Average Temperature DGSR	ARMAX LSTM RF SVR	Daily	MSE: 0.8645 RMSE: 0.9298 MAE: 0.7033 MAPE: 0.9545 MSPE: 5.4402 R2: 0.7751

only highlights the current state of research but also points to areas ripe for future exploration, driving the field towards more precise and varied applications of transformer technology in solar energy forecasting.

3.1. *Very short-term solar forecasting*

In the domain of very short-term solar forecasting (ranging from 1-min to 1-h), an intricate analysis of the reviewed models elucidates a multifaceted landscape where each approach's performance is a result of its unique technological and methodological advancements. Here, we delve into the quantitative nuances that these models exhibit, offering a comparative synthesis that highlights correlations and divergences in their predictive prowess. In this leu, Fig. 11 showcases MAE visualization for some models as all the authors did not follow the same metrices for comparison.

Starting with Liu et al. [166], their transformative approach integrating Informer and VIT models is distinguished by MAE values ranging from 34.21 to 49.53 W/m^2 for 10-min forecasts. The performance leap in this model can be attributed to the dynamic feature encoder based on Vision Transformer, which adeptly captures global features from sky image sequences. This is a methodological enhancement over traditional convolutional methods, which often focus on local perspectives. Moreover, the inclusion of a cross-modality attention mechanism elucidates the model's ability to fuse disparate modalities efficiently, contributing to its robust performance across validated horizons.

Gorantla et al. [95] achieve an nMAPE of less than 10% and an R2 of 0.85 using ViTs pre-trained on ImageNet-21 k. The pre-training on a large-scale dataset like ImageNet-21 k lends the model a generalizability that allows it to maintain performance even with reduced training data, a testament to the efficacy of transfer learning in enhancing adaptability to different geographical environments.

The Multivariate Transformer model [170] presents an MSE of 0.3422, surpassing traditional LSTM and GRU methods. The model's strength lies in its capacity for very short-term forecasting which is

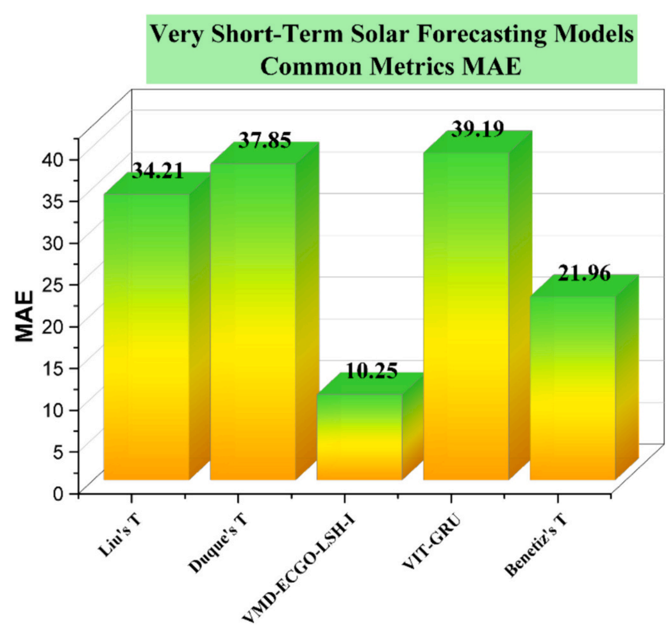

Fig. 11. MAE values of very short-term solar forecasting models.

invaluable in contexts like energy trading. This quantitative edge is likely a consequence of the model's multivariate nature, considering multiple input parameters to refine its predictive accuracy. Duque et al. [72] research further consolidates the supremacy of Transformer models in forecasting GHI over a full day with 15-min intervals, achieving a MAE of 37.85 W/m^2 and an RMSE of 72.24 W/m^2.

Zhang et al.'s [171] adoption of Transformers combined with gating mechanisms, the SIAT model [110] utilizing a GPT-2 based decoder, and the Multi-Branch ResNet-Transformer [101] each offer distinct methodological novelties. Zhang et al. [171] improve balanced accuracy for ramp events by 9.43% and 3.91%, indicating a nuanced handling of rapid solar intensity shifts. SIAT's [110] proficiency in assimilating temporal and spatial data from image sequences results in an RMSE of 112 W/m^2 for 15-min predictions, while the Multi-Branch ResNet-Transformer demonstrates superior spatio-temporal forecasting accuracy.

In juxtaposing the above models, it becomes evident that methodological innovations such as cross-modality attention, transfer learning, and multi-branch architectures confer significant predictive advantages. Each model's quantitative performance is enhanced by these technological advancements, which are tailored to address the specific complexities of SI data.

Furthermore, when examining models such as the VMD-ECGO-LSH-Informer [156] and CT-NET [133], there's a discernible trend towards incorporating hybrid architectures for a more comprehensive understanding of solar data. The hybrid model of VIT-GRU [165] is a notable contributor in 5, 10, and 15-min forecasting horizons. With MAE improvements such as 39.19, it demonstrates exceptional performance under diverse weather conditions, highlighting its potential for reliable grid operation. Additionally, the VIT-Based Model of Kang et al. [120], utilizing auxiliary sensor information, excels in longer prediction times, showcasing its advanced cloud movement anticipation capabilities, particularly over 300 s and 600 s intervals.

The VMD-ECGO-LSH-Informer [156] model, for instance, improves MSE values by up to 59.89% for 5-min intervals through an intricate combination of signal decomposition and enhanced chaos game optimization algorithm, underpinning the interplay between data preprocessing and algorithmic refinement in forecasting. Contrastingly, the CT-NET [133] model leverages a CNN-Transformer architecture, yielding a low MSE of 0.0414, which underscores the synergetic potential of convolutional layers for local feature extraction and transformers for contextual understanding. This hybridity not only lowers the error rates but also suggests a lightweight and efficient model suitable for smart grid integration.

These models underscore a significant relationship between enhanced algorithmic structures and predictive performance. The correlation lies in the degree to which each model harnesses spatial, temporal, and even cross-modality features – often translated into lower MAE and RMSE values and higher R^2 scores. The advancements in convolutional layers, attention mechanisms, and multi-modal learning frameworks present in these models are directly responsible for the quantitative leaps in forecasting accuracy.

3.2. Short-term solar forecasting

The field of SI/GHI forecasting has witnessed a paradigm shift with the integration of advanced deep learning models, each bringing a unique dimension of innovation and enhancing the accuracy and adaptability of predictions across various time horizons especially Short-Term forecasting normally referred as hour ahead or day ahead forecasting as lies within the time range of 1-h to 24-h.

3.2.1. SI/GHI forecasting

Recent advancements in SI forecasting have been influenced by both technological and methodological innovations. Fig. 12 showcase the common metrics of models which predicts SI/GHI. A common thread across successful models such as TFT [147] and ProSIT [108] is the effective handling of multi-step forecasting and probabilistic outcomes, suggesting a focus on temporal dynamics and uncertainty quantification in solar data.

The TFT [147] model's multi-step forecasting capability for GHI is a notable technological evolution. Its adaptability across various climatic conditions, with forecast skill scores fluctuating between 0.3 and 0.7, indicates sophisticated handling of temporal dependencies and external

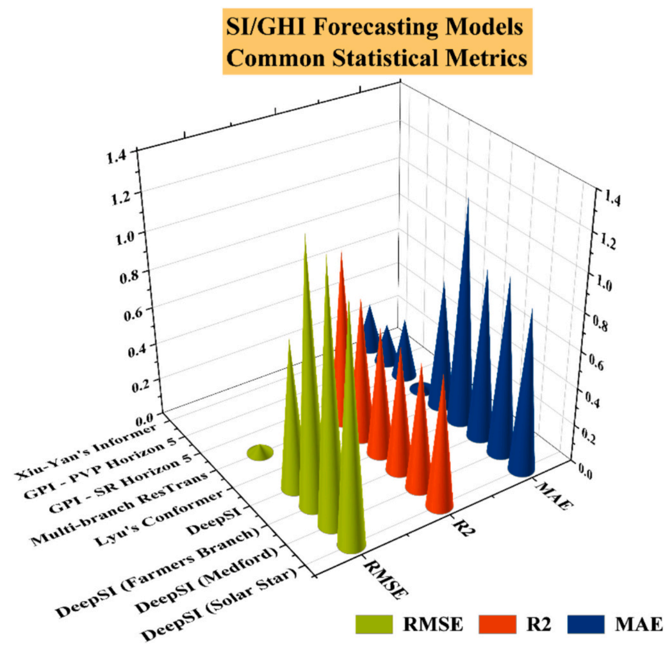

Fig. 12. Common statistical metrics of Short-term SI/GHI forecasting models.

variables. Similarly, the ProSIT [108] model's superior performance in probabilistic forecasting, with a reduced RMSE and improved q-risk scores, exemplifies the integration of advanced statistical methods to handle uncertainty, a critical factor given the stochastic nature of solar data.

The integration of Aerosol Optical Depth (AOD) in the Informer model [83], yielding an MSE of 0.156 and MAE of 0.262, underscores the methodological refinement of incorporating environmental factors directly affecting GHI. This shows a clear link between model robustness and the inclusion of specific atmospheric parameters that influence SI. Furthermore, the self-attentive transformer [80] architecture, achieving significant reductions in forecasting error rates, illustrates the technological shift towards architectures capable of capturing complex nonlinear trends in SI data. The RSAM [109] model leverages this architecture for multi-horizon forecasting, offering superior accuracy evidenced by RMSE values of 52.45 and 64.87 for the India and Hotevilla datasets, respectively. This suggests that attention mechanisms in deep learning architectures enhance model sensitivity to diverse climatic conditions.

The advancement represented by the DeepSI [145,146] model, designed to predict daily SI using downscaled climate variables, highlights the role of hybrid models that combine LSTM autoencoders and transformers. Its performance, indicated by lower RMSE and higher R2 scores compared to traditional methods, showcases an effective methodological advance in long-term irradiance projections, likely owing to the model's ability to learn from large historical climate datasets and its suitability across various geographic locations. The emergence of transformers in models like RSAM [109] for multi-step forecasting demonstrates an evolving landscape where non-recurrent, parallel processing models handle large datasets more effectively. The enhanced performance of RSAM [109], especially for longer prediction horizons, validates the technological advantage of self-attention mechanisms over traditional sequential models.

In summary, the deep analysis of the transformer's models for SI/GHI shows that the performance enhancements in solar forecasting models are largely due to the integration of environmental parameters such as AOD [83], the adoption of hybrid models [108,145–147] for long-term forecasting, and the application of self-attention mechanisms [80] for complex pattern recognition across various time horizons. The quantitative assessment reveals that these technological and methodological advancements collectively facilitate significant improvements in forecasting accuracy, robustness against atmospheric disturbances, and adaptability to different geographic and climatic conditions.

3.2.2. PV power forecasting

The progression in solar power forecasting, as exhibited by the ensemble of models delineated in the compilation, is a testimony to both incremental and leapfrog advancements in the domain as showcased in Fig. 13. Delving into the model characterized by hybrid CNN-LSTM-Transformer architecture [132], one observes a fusion of CNNs [172] with LSTM [173] layers, effectively amalgamated into the Transformer's architecture. The convolutional layers empower the model with a robust spatial feature detection capability, while LSTM layers manage temporal dependencies—both crucial for the predictive task at hand. The integration within a Transformer framework, known for its self-attention mechanism, enables the model to discern intricate patterns over extended time lags, which traditional sequence models might overlook. This methodological confluence is quantitatively manifested in the model's empirical performance, with commendable RMSE (0.344), MAPE (0.041), and MAE (0.393) metrics.

Similarly, the innovation brought forth by the H-Transformer [144] synthesizes SARIMA's [174] linear forecasting excellence with the Transformer's adeptness at handling nonlinear dependencies. The methodological innovation here lies in effectively leveraging the Transformer to model residuals from the SARIMA output, enhancing predictive capabilities and exhibiting versatility as evidenced by lowered RMSE (766 & 1307) and MAE (516 & 934) across datasets. Following the similar architecture, IST-LSTM-Informer by Cao et al. [136], employing a stacking ensemble algorithm, leveraged LSTM and Informer to enhance short-term and medium-term PV power forecasting, demonstrating the strengths of combining different methodologies.

The integration of graph convolutional layers in models like GCLSTM [153] and GCTrafo [153] marks a methodological shift towards capturing spatial dependencies within PV production data. This represents a tailored approach for multi-site forecasting scenarios, offering a substantial leap in addressing spatial relationships within data, pivotal for the accuracy of such models. The fusion approach is also visible in models like MATNet [85] and TFT [148], where a blend of deep learning and physical-based modeling enhances day-ahead forecasting accuracy.

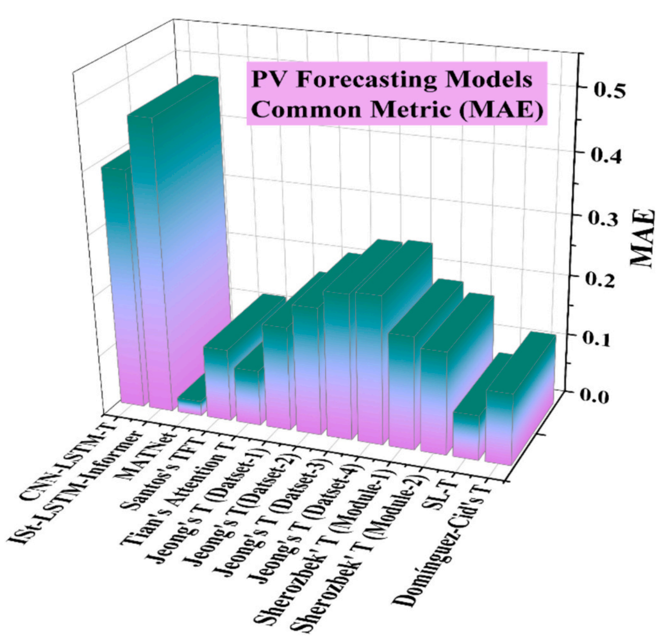

Fig. 13. MAE of Short-term PV forecasting models.

MATNet's [85] employment of self-attention mechanisms allows dynamic focus on relevant input data elements, which, when combined with physical model insights, results in highly accurate forecasts, as indicated by the model's MSE (0.0027) and MAE (0.0245).

Furthermore, the strategic incorporation of XGBoost [175] for feature selection in Transformer-based models is a methodological enhancement that improves the predictive model's focus by identifying and emphasizing salient features. This is particularly evident in the case of the DTTrans [137] model, where the incorporation of Delaunay triangulation [176] and TransGRU [177] signifies an advancement over conventional models, as it robustly handles weather forecast uncertainties achieving NMAE of 10%. Lastly, Transformer-LUBE [162] marks a significant stride in probabilistic solar power forecasting. By combining pre-processing and post-processing techniques, this model achieves lower NMAPE and NRMSE values, highlighting the importance of comprehensive data handling in forecasting.

Collectively, these case studies exemplify a methodological sophistication in the field. Enhanced data preprocessing, for instance, the employment of clustering techniques for seasonal adjustment, has enabled models to adeptly factor in temporal variabilities intrinsic to SI. The integration of hybrid modeling techniques has facilitated a leveraged synthesis of disparate algorithmic strengths, yielding a compound effect on predictive precision. Notably, graph-based models capture spatial interdependencies inherent to PV output, an imperative for forecasting across multiple geolocations.

XGBoost's [175] role in feature optimization streamlines the input feature space, thereby enhancing subsequent deep learning model performance. Its deployment in the Transformer-based [137] framework allows for a selective focus on salient features, indicative of a strategic evolution in data handling techniques. The integration of Transformer models within the solar forecasting architecture heralds a significant advancement, leveraging the model's innate strength in capturing long-range temporal dependencies and complex patterns within sequential data.

Each technological and methodological enhancement contributes uniquely to the precision of solar power forecasting models. For instance, the incorporation of LSTM [173] layers within the CNN-Transformer [132] architecture reflects a bidirectional enhancement in temporal analysis, while the usage of clustering techniques for pre-processing exemplifies a nuanced understanding of feature relevance tied to seasonal cycles. Collectively, these advancements illustrate an overarching trend towards a more nuanced, intricate, and ultimately precise understanding of solar power generation patterns, facilitating more accurate and reliable predictive models that can better integrate into energy management systems.

3.2.3. Time series forecasting (solar data set)

In the realm of time series forecasting, tailored to solar datasets, the emergence of novel model architectures has notably advanced predictive accuracy, specifically for SI and solar power generation forecasting.

Probabilistic Decomposition Transformer (PDTrans) [105,106], leveraging a hierarchical approach augmented by probabilistic decomposition [178], showcases its aptitude in interpreting complex temporal patterns, particularly the pivotal periodicities that mark solar data. This model exhibits proficiency in hourly solar dataset forecasting, distinguished by its performance metrics $\rho0.5$ and $\rho0.9$ at 0.205 and 0.073, respectively. PDTrans [105,106] elevates the standard for time series forecasting, melding Transformer architecture with conditional generative models [179] to provide forecasts of superior detail and interpretability. Fig. 14 presents a comparative view of PDTrans with other similar models.

GQFormer [107] presents a multivariate forecasting model that estimates multiple quantiles [180], broadening the forecasting scope for solar datasets. Despite its limitations, the model contributes diversity to the forecasting landscape achieving $QL_{0.5} = 21$ and $QL_{0.9} = 8.2$ when compared with TRMF [181], DeepAR [182] and LogTrans [183]. AST

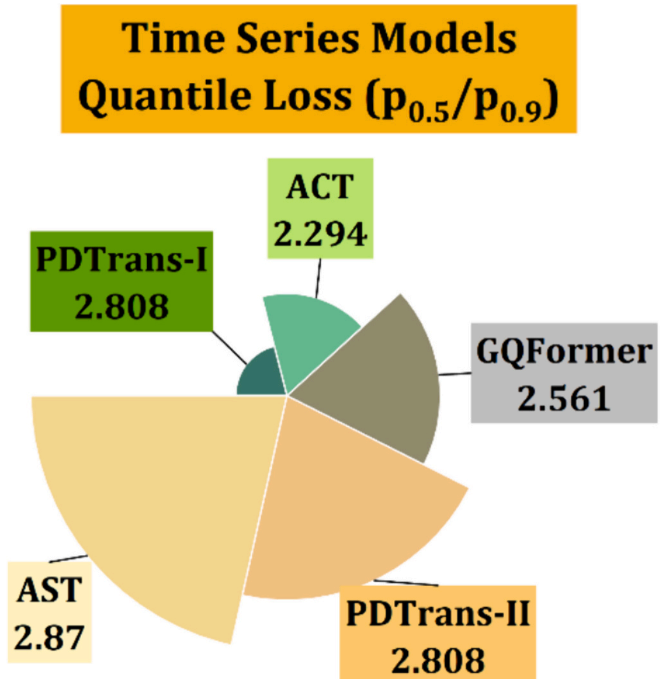

Fig. 14. Quantile Loss of Time Series forecasting models.

[87], an Adversarial Sparse Transformer, distinguishes itself by integrating sparse attention mechanisms and adversarial training [184], significantly enhancing accuracy for solar datasets with notable Q50/Q90 loss values of 0.155 and 0.054 on an hourly scale, highlighting its utility for long-horizon forecasting.

iTransformer [98] makes strides with an inverted attention mechanism [185] that amplifies its forecasting capability by 38.9% compared to conventional Transformers, illustrating the potential of re-envisioned architectures in forecasting. ACT [86], with its one-step decoding and convolutional attention blocks [186], adeptly manages intricate short-term and long-term patterns within solar datasets achieving Q50/Q90 = 0.195/0.085, demonstrating versatility across various forecasting granularities. SpringNet's [134] innovative approach leverages Spring DTW attention layers, focusing directly on time series shape patterns [187], particularly in solar power forecasting. Achieving the finest MAE (0.258 and 0.358) and RMSE (0.380 and 0.494) metrics, SpringNet [134] has markedly expedited the model training process and showcased unparalleled accuracy in capturing and forecasting solar power generation in intervals from 30 min to an hour.

These models each contribute uniquely to the progression of solar dataset forecasting. Their diverse methodologies, from hierarchical probabilistic approaches to attention-based mechanisms, offer comprehensive solutions for predicting solar data. Their capacity to handle various time horizons and deliver precise forecasts is paramount to solar energy management and predictive analytics, underpinning the evolution of the renewable energy sector and enhancing the efficacy of solar energy production and consumption systems.

3.2.4. *Evaluating significant input parameters*

Jiang et al. [82] examined the effectiveness of the Informer model in forecasting global solar radiation using datasets from various climatic zones. They emphasized the critical role of choosing optimal input parameters. It was found that Informer performed exceptionally well when using a combination of the clearness index and pressure as inputs. Compared to other reference models such as Random Forest, Extra Trees [188], Bagging[189], Decision Tree[190], and XGBoost, Informer showed significant improvements in predictive performance. Specifically, the improvement in R2 was observed in the range of 20% to 75%, and reductions in RMSE by more than 95% under Informer's optimal inputs were reported. This illustrates the model's ability to handle multiple inputs efficiently, as the computational demand remains relatively stable even when inputs increase from 2 to 4. However, Jiang et al. [82] also revealed that univariate inputs result in inferior prediction accuracy, indicating a limitation of the Informer model in processing single-parameter inputs. Furthermore, it was highlighted that increasing the sampling frequency in long sequence predictions does not necessarily lead to a noticeable improvement in accuracy. For instance, with higher sampling frequencies, the computational time of the model increases significantly, but the increase in accuracy is minimal and, in some cases, even a decrease occurs. This finding suggests that enhancing prediction accuracy through increased sampling frequency may not be a viable strategy in long sequence forecasting with the Informer model.

3.3. *Medium-term forecasting*

The comprehensive analysis of Sinha et al.'s [111,112] research on sub-seasonal and week-ahead SI forecasting using deep sequence learning presents a significant advancement in solar energy prediction. These investigations are instrumental in developing ML models that enable precise point and probabilistic predictions for SI, leveraging the SURFRAD dataset across various U.S. stations in 2018.

In the sub seasonal forecasting exploration [112], the study endeavors to predict SI for an approximate lead time of two weeks (168 h), harnessing an assortment of deep learning methodologies including Temporal CNN (TCN) [191], TCN with attention, Transformers, Ngboost [192], and LSTM. These models are bench-marked against Smart Persistence (SP) [193] and Hourly Climatology [194]. For probabilistic forecasts, the Continuous Ranked Probability Score (CRPS) [195] was employed as an evaluative metric. The Transformer model particularly excelled, outshining Ngboost [192] and LSTM in point predictions. It displayed slightly lesser prowess in probabilistic forecasting, yet still presented robust performance against the CH-PeEN benchmark [196] with CRPS scores at 77.69 and marked RMSE improvements over SP and Hourly Climatology by 32.95% and 19.4%, respectively.

The week-ahead forecasting [111] endeavor extends the previous study's findings, illustrating that deep sequence learning algorithms have considerable potential for generating enhanced forecasts with a lead time of one week. Here, models such as LSTM, TCN, TCN with Attention, and Transformers generally surpassed benchmarks like SP, Hourly Climatology, and Numerical Weather Prediction (NWP) models [194] in point forecasts. With respect to probabilistic forecasting, TCN and TCN with Attention displayed exceptional performance, whereas Transformers did not exhibit the same level of proficiency, barring their results at the Desert Rock station where RMSE was noted at 131.04.

Collectively, these studies [111,112] underscore the efficacy of integrating ML techniques into the framework of long-term solar power forecasting. The strategic inclusion of comprehensive input variables, from Global Horizontal Irradiance (GHI), solar zenith angle, to a suite of meteorological data, significantly contributed to the models' robustness. This endeavor is crucial for the effective operation of power systems and the management of renewable energy resources. The findings underscore the promise of deep learning for sub seasonal SI forecasting.

3.4. *Unspecified forecasting time research*

The studies conducted by Kim et al. [130], Benítez et al. [78], and Luis et al. [119] have underscored the transformative impact of Transformer-based models in the domain of time series forecasting and solar PV profiling. Nonetheless, a comprehensive analysis of these studies reveals a necessity for a more rigorous examination of the factors contributing to performance enhancements in each model.

For instance, the Convolutional Transformer Model presented by Kim et al. [130] integrates convolutional layers [197] with Transformer architecture, which may suggest an inherent synergy between localized feature extraction and global dependency modeling. The study indicates a proficiency of the model with RMSE, RRSE and CORR being 0.269, 0.273 and 0.748 respectively in deciphering datasets with closely correlated variables, a capacity which merits deeper inquiry into the specific model components responsible for such nuanced interpretations.

In the evaluation by Benítez et al. [78], the Transformer architecture was juxtaposed against LSTM [173] and CNN models [172], showcasing comparative superiority with WAPE (13.550) and MAE (2.246) in certain datasets. The application of a Multi-Input Multi-Output (MIMO) [198] strategy and the preprocessing steps warrant a thorough analysis to discern their roles in achieving the stated forecasting accuracy, as indicated by the WAPE and MAE metrics for the solar dataset.

Luis et al. [119] introduced SolarFormer, harnessing a multi-scale Transformer [199] encoder and a masked-attention Transformer [200] decoder, tailored for solar PV profiling. The significant gains in performance metrics (Intersection over Union (IoU) of 88.78, F1 score of 94.05, and Accuracy of 94.39%) on various datasets prompt a critical appraisal of the model's design. It is essential to delve into the multi-scale approach and the masked-attention mechanism's efficacy in enhancing solar PV understanding and addressing homogeneity challenges within the datasets.

To encapsulate, these studies demonstrate the adeptness of Transformer models across a variety of applications, from multivariate time series prediction to solar PV profiling. However, the omission of precise forecasting times across these investigations precludes a comprehensive understanding of the models' temporal applicability in pragmatic scenarios. This absence delineates a lacuna in the full apprehension of how these models may be deployed within real-world temporal constraints.

4. Computational efficiency in solar forecasting: Bridging innovation and practicality through transformer models

The pursuit of computational efficiency in solar forecasting models represents a critical intersection of innovation and necessity, with significant implications for energy management and grid integration [10]. This exploration, grounded in the examination of nearly seventy scholarly articles, charts the progression from single model to hybrid frameworks, offering insights into the computational strategies pivotal for the future of solar forecasting. Computational efficiency, characterized by the judicious use of time, memory, and processing power, is not merely a technical consideration but a crucial factor for real-world applications [201]. It determines a model's viability in delivering timely and accurate forecasts, essential for the seamless integration of renewable energy sources into the power grid [202]. The review highlights the crucial balance models must strike between forecasting accuracy and computational demands, a balance that becomes increasingly important in scenarios of limited computational resources. This balance, and the strategic model selection it informs, is paramount for energy grid operators and stakeholders in navigating the challenges of grid stability and efficient energy distribution [203].

4.1. Single model frameworks: Unveiling efficiency and efficacy

4.1.1. Pioneering with transformers

The advent of Transformer models has marked a revolutionary shift in solar forecasting, introducing a level of computational efficiency [10] and data processing capability previously unattainable. The Constrained Transformer Model [66] is designed with a selective attention mechanism that enhances real-time responsiveness by focusing only on significant features. This approach not only accelerates processing but also ensures accuracy in dynamic SI environments. Similarly, the Attention-based Transformer Model [77] incorporates an interpretable mechanism that visualizes the impact of specific features on predictions, promoting transparency and facilitating deeper insights into the decision-making process, which is crucial for model validation and refinement.

In the realm of multimodal data handling, the Transformer-based Multimodal Learning Framework [166] utilizes separate transformers for each data modality before combining their outputs. This early integration ensures that intrinsic patterns from each type of data are effectively captured, boosting the model's overall predictive performance. Adversarially trained models like the ACT [86] and AST [87] integrate adversarial robustness with a focus on local and global data dependencies. ACT [86] employs convolutional self-attention to incorporate local context processing, traditionally a strength of CNNs, while AST [87] uses a sparse attention setup to enhance focus and computational efficiency.

The TFT [148] stands out by integrating gating mechanisms that selectively process information across time steps, optimizing computational resources and improving model interpretability in handling complex datasets. In addition, the Transformer-LUBE model [162] introduces probabilistic forecasting to the transformer domain, providing estimates of confidence intervals around predictions through the Least Uncertainty-Based Estimation method, crucial for risk-sensitive environments. Focusing on short-term forecasting, the Transformer-based model [162] for Short-term PV Generation Forecasting tailors the standard transformer architecture to address the specific variabilities of solar generation data, enhancing its applicability in operational settings.

The Informer model [82] advances this concept with a 'ProbSparse' self-attention mechanism, significantly reducing computational complexity and making long-sequence forecasting more efficient. Customizing this approach, the SI Prediction model [204] adapts the self-attention mechanism to emphasize meteorological factors impacting solar outputs, optimizing the transformer's performance for weather-sensitive predictions.

Expanding the scope to spatial variability, the SolarFormer [119] uses a multi-scale transformer architecture to process data at various resolutions, thereby capturing fluctuations in PV output effectively across different conditions and scales. The integration of graph neural networks with transformers in the Spatio-Temporal Graph Neural Networks model [153] for Multi-Site PV Power Forecasting enhances its capability to handle complex interactions between multiple sites.

For applications involving image data, the Vision Transformer-Based PV prediction model [120] employs a visionary approach to encode spatial relationships within images, crucial for utilizing satellite and sky imagery effectively in solar forecasting. This model demonstrates the transformative potential of applying vision transformer techniques to the field of SI prediction.

4.1.2. Evolving with CNNs and LSTMs

The evolution of solar forecasting has also seen substantial contributions from CNNs and LSTMs. The Convolutional Transformer Model [130] for Multivariate Time Series prediction innovatively combines the local receptive fields of CNNs with the global context sensitivity of transformers. This hybrid architecture allows the model to efficiently manage multivariate time series data by capturing both spatial and temporal dependencies, making it highly suitable for complex datasets found in SI forecasting.

Similarly, the Dual-Encoder Transformer [94] for Short-Term PV power prediction employs two parallel encoders—one for temporal PV data and another for contextual environmental factors. This dual-encoding structure optimizes the model's performance by simultaneously processing and learning the distinct but interrelated patterns between environmental conditions and solar power outputs.

Deep Learning-Enabled Prediction [145] of daily SI from simulated climate data and the DeepSI Model [146] both highlight the power of deep learning in forecasting SI with high precision. The former [145] utilizes simulated climate data to train a model that can generalize well to real-world conditions, while DeepSI [146] incorporates advanced techniques such as Monte Carlo dropout for uncertainty quantification, ensuring robustness and reliability in its predictions.

DTTrans [107] introduces a novel approach by incorporating geometric triangulation into the forecasting model. This allows the model to optimize spatial relationships and improve the accuracy of PV power predictions by strategically leveraging data from triangularly positioned weather stations around the target site.

Solar Energy Production Forecasting Based on a Hybrid CNN-LSTM-Transformer model [132] effectively utilizes the strengths of CNNs, LSTMs, and transformers. This comprehensive model harnesses CNNs for extracting spatial features, LSTMs for capturing temporal sequences, and transformers for integrating these features over long sequences. This synergy facilitates a deeper understanding of both immediate and contextual influences on solar production.

Lastly, Short-term SI Prediction from Sky Images with a Clear Sky model [117] uses a cutting-edge approach by integrating sky image analysis directly into the forecasting model. This integration allows for the direct use of visual data, enhancing the model's predictive accuracy by including real-time observations of cloud patterns and other atmospheric conditions.

4.2. The rise of hybrid model frameworks: Bridging computational strategies

As solar forecasting tasks grow in complexity, hybrid model frameworks have emerged, synthesizing multiple computational strategies to achieve a delicate balance between efficiency and performance.

4.2.1. Synergizing models

The Synergizing models reflects a sophisticated merging of multiple advanced techniques to enhance the predictive capabilities and computational efficiency of solar and wind power forecasting systems. Each model brings together different methodologies, demonstrating the power of hybrid approaches in tackling the intricacies of renewable energy data.

The Transformer with multi-head attention and feature selection using Pearson Correlation Coefficient model [71] exemplifies the effective combination of feature selection with advanced neural architectures. By incorporating Pearson Correlation Coefficients for feature selection, the model efficiently identifies and prioritizes the most relevant features before processing them through a multi-head attention mechanism. This pre-processing step ensures that the transformer focuses its capacity on the most impactful data, optimizing both accuracy and computational resources.

The Enhanced Chaos Game Optimization Algorithm and Locality Sensitive Hashing Based Informer Model [156] integrates innovative algorithms to improve the model's ability to handle high-dimensional data spaces. The Chaos Game Optimization algorithm enhances the global search capabilities, while Locality Sensitive Hashing streamlines the handling of large-scale data by reducing dimensionality, which, when combined with the Informer architecture, provides a robust framework for capturing long-term dependencies with increased efficiency.

The Multi-Head Attention (MHA) utilized model [164] for medium-term PV and Wind power prediction leverages the MHA mechanism to parse multiple time horizons within a single framework, allowing the model to simultaneously learn from various temporal scales. This multi-scale approach is crucial for medium-term forecasting where the integration of short-term weather fluctuations and long-term seasonal trends are vital for accuracy.

Multi-timescale PV power forecasting using an improved Stacking Ensemble Algorithm based LSTM-Informer model [136] showcases the fusion of LSTM and Informer models through a sophisticated stacking ensemble method. This model benefits from LSTM's ability to capture short to medium-term dependencies and the Informer's efficiency in processing long sequence data. The ensemble approach enhances model robustness by aggregating diverse predictive insights, which is particularly effective across multiple timescales.

The TEFNEN [143] is tailored for the complexities of natural environmental data, employing a transformer-based model that adapts to the high variability and non-linear characteristics of such datasets. TEFNEN's architecture is designed to handle erratic data patterns typical in natural environments, making it well-suited for real-world applications in renewable energy forecasting.

Lastly, the Transformer-based hybrid forecasting model [144] for multivariate renewable energy combines transformers with other neural network models to handle the multivariate aspects of renewable energy systems. This model utilizes a hybrid approach to effectively process and integrate various data types, from meteorological to operational parameters, ensuring comprehensive analysis and predictive performance.

4.2.2. Fusing deep learning with traditional forecasting

In the domain of "Fusing Deep Learning with Traditional Forecasting," the models represent a blend of cutting-edge deep learning architectures with established forecasting techniques, aimed at enhancing both predictive accuracy and interpretability in solar power forecasting.

The Hybrid Models [111] combining Temporal CNNs, Temporal CNNs with attention, and Transformer capitalizes on the strengths of each component to handle complex time series data effectively. Temporal CNNs are adept at extracting localized patterns in time series data, and when enhanced with attention mechanisms, they provide a focused analysis of relevant temporal features. The inclusion of the Transformer architecture allows for the integration of these features across longer time sequences, offering a comprehensive view of both immediate and extended temporal dynamics. This hybrid approach ensures that the model not only captures detailed temporal patterns but also contextualizes them in a broader sequence, enhancing both the precision and relevance of its forecasts.

Sub seasonal solar power forecasting via Deep Sequence Learning [112] leverages deep learning techniques to predict solar power output over sub seasonal timescales. This model employs a deep sequence learning framework that combines multiple layers of neural networks to process and learn from complex patterns in solar data over extended periods. The model's strength lies in its ability to synthesize vast amounts of data and extract meaningful patterns for long-term forecasting, providing crucial insights for energy management and planning over sub seasonal periods.

Hybrid approaches such as the CT-NET [133] combine the strengths of CNNs for local feature extraction with transformers' ability to learn long-range dependencies. This makes it exceptionally adept at handling spatial-temporal patterns crucial in PV power forecasting. The Transformer and Spring DTW [134] for Time Series Forecasting introduces an innovative fusion of transformer technology with the traditional DTW method. Dynamic Time Warping is a technique used to measure similarity between two temporal sequences which may vary in speed. By integrating DTW with transformers, the model gains the ability to align sequences in a nonlinear fashion, accommodating variations in timing and duration within the data. This capability is particularly useful in solar forecasting, where SI patterns can shift significantly due to environmental factors.

4.3. Navigating the trade-offs: A path forward

The exploration of single and hybrid model frameworks in solar forecasting unveils a dynamic interplay between computational efficiency and forecasting efficacy. While single model frameworks continue to offer significant advancements in data processing and accuracy, the emergence of hybrid models represents a paradigmatic shift towards versatile, computationally efficient forecasting solutions. These models not only encapsulate the strengths of individual computational strategies but also present a holistic approach to addressing the multifaceted challenges of solar forecasting.

In navigating these trade-offs, the choice between single and hybrid frameworks should be informed by the specific needs of the application at hand, considering factors such as computational resource availability, the immediacy of forecasting requirements, and the desired level of accuracy. This nuanced selection process underscores the importance of ongoing innovation and adaptation in the field of solar forecasting, ensuring that as computational methodologies evolve, they continue to serve the pressing needs of energy management and sustainability in the face of a changing climate.

5. Overview of model outcomes and hyperparameters

So far, there have been 64 models employed to predict SI and PV Generation. These models possess varying degrees of efficacy based on their unique configurations and hyperparameters, available as Appendix-2, a snippet of which is presented as Table 2. Although every model is trained for specific dataset and region, yet an effort is made to analyze the hyperparameters in this section using three different approaches. Firstly, a hyperparameter analysis is done on single and hybrid models on the basis of their statistical metrices. Secondly, the model outcome like the predicted output is analyzed on the basis of various hyperparameters. And thirdly the importance of various hyperparameters is highlighted briefly.

5.1. Hyperparameters analysis on model type

5.1.1. Single models

The landscape of single Transformer models showcases a spectrum of architectures tailored for diverse applications, with hyperparameters playing a pivotal role in their success. The MATNet [85] model stands as a prime example, achieving remarkable accuracy with an MSE of 0.0027, RMSE of 0.0460, and MAE of 0.0245. This precision is attributed to its strategic configuration, including a 512-dimensional model, 8 parallel heads, and 3 sub-encoder layers, optimized for a 24-h horizon. This setup not only demonstrates the impact of model dimensionality and attention mechanisms on prediction accuracy but also aligns with findings from Aleissaee et al. [205], who underscore the significance of such hyperparameters in the realm of remote sensing, enhancing model performance across various tasks.

Further illustrating the adaptability and effectiveness of Transformer models, the work by Sherozbek et al. [206] presents models achieving RMSEs of 0.24 and 0.21 for different modules, despite the absence of detailed hyperparameter specifics. Similarly, the Vision Transformer (VIT) [120] has proven its effectiveness in image-based forecasting, achieving commendable MSE and MAPE values with a learning rate of 0.0001 and the Adam optimizer [207], showcasing the importance of learning rates and optimization strategies.

Kim et al.'s [208] Transformer, with its impressive PCC of 0.999 and R2 of 0.997, leverages 2 encoder and decoder layers with 8 attention headers, emphasizing the critical nature of encoder-decoder layer configuration in capturing temporal data patterns. This importance is echoed in the multimodal learning survey by Xu et al.[209] which highlights the indispensability of hyperparameter optimization for enhancing performance across a variety of learning applications.

Cid et al.'s [143] model, specialized in PV Generation, achieves an MAE of 0.11827 and an RMSE of 0.29098 for power, utilizing a 24*5-h encoder sequence and a 24-h decoder sequence with 2 attention heads and 2 layers for both encoder and decoder. This model further highlights the critical role of hyperparameters, including attention heads, encoder-decoder layers, and model dimensionality, in enhancing forecasting accuracy—a sentiment reinforced by the systematic review from Faraone et al.[210], which points to hyperparameter optimizations as significantly associated with increased model performance in genomic machine learning research.

These models collectively underscore the nuanced yet critical role of hyperparameters in defining and enhancing the accuracy and adaptability of Transformer models across various domains, from remote sensing and image forecasting to genomic research, aligning with contemporary evidence and systematic reviews.

5.1.2. Hybrid models

Hybrid models, by design, amalgamate distinct neural network architectures to harness their individual strengths, thus enhancing model performance. For example, the ISt-LSTM-Informer [136] model illustrates this concept by combining LSTM units of 100 and 120 with an Informer d_model of 512, achieving a broad spectrum of performance (RMSE from 0.617 to 2.285 and MAE from 0.478 to 0.997). This model's adaptability and accuracy underscore the utility of integrating different architectures, a principle supported by Maglaveras et al. [211]. Similarly, the CNN-LSTM-Transformer [132] model combines a CNN with two convolution layers, an LSTM layer of 32 units, and a Transformer with four encoder and two decoder layers, exemplifies the capture of both spatial and temporal dynamics, achieving impressive metrics (RMSE of 0.344, MAPE of 0.041, and MAE of 0.393). The integration of CNN and LSTM layers, alongside Transformer encoders and decoders, aligns with the observations by Yu et al. [212].

The adaptability of hybrid models like the H-Transformer [144] across different scenarios, particularly in solar dataset prediction, highlights the importance of a wide range of hyperparameters including units from 4 to 256 and learning rates from 0.1 to 0.001. This adaptability is further emphasized in the systematic review by Azad et al. [213], which explores the applications of Transformers in medical imaging, demonstrating Transformers' capability to learn long-range dependencies and spatial correlations, thereby offering a significant advantage over traditional CNNs in complex visual tasks.

Specialized models also exhibit significant results, with GCTrafo [153] model indicating NRMSE of 10.83 and NMAE of 29.89, using a 1D-convolutional kernel size of 4 and 8 attention heads. This model's performance underscores the importance of kernel size and attention heads in handling graph-based data. Constrained Transformer [66] gains R2 of 70.4%, MAE of 11.6 KWh, and RMSE of 6.7kWh, showcasing solid performance in PV Generation forecasting. Attention Transformer [67] shows MAE of 0.092 and MSE of 0.036, configured with 40 nodes and 100 iterations, highlighting its precision in complex data handling [214].

The efficacy of these models is largely influenced by their hyperparameters. For instance, Transformer models with higher-dimensional

Table 2
Hyperparameters of Reviewed Models.

Hyperparameter	Range of Values/Description
Liu et al. Hybrid (Informer+Transformer) [166]	
Training epochs	10
Early-stopping (patience)	3
Total space complexity	Represented by floating point operations (FLOPs)
Temporal consumption	52.0% for Informer compared to Reformer
LSTM training time	291.2 s/epoch
LSTM total time	3560.2 s
Gorantla et al. Vision Transformer (ViT) [95]	
Number of layers	1 to 7
Batch size	512
Learning rate	0.001
Weight decay	0.1
Optimizer	Adam
Loss function	Mean Squared Error (MSE)
Yan et al. Informer [83]	
Encoder Input	1 × 3Conv1d Embedding (d = 512)
Attention Heads	Multi-head ProbSparse (h = 16, d = 32)
Dropout Probability	0.1
FeedForward Network Inner Dimension	2048
Convolution	1 × 3 conv1d, ELU
Pooling	Max pooling (stride = 2)

models and multiple attention heads, e.g., MATNet [85] and Kim et al.'s [208] Transformer, tend to show lower error rates and higher accuracy. The hybrid models, such as ISt-LSTM-Informer [136] and CNN-LSTM-Transformer [132], demonstrate their strength in integrating different architectures to capture both spatial and temporal dynamics, as reflected in their varying RMSE and MAE.

In contrast, single and specialized models like GCTrafo [153] and Constrained Transformer [66], with their unique configurations, cater to specific forecasting tasks, leading to tailored predictions for distinct datasets. This is evident in their NRMSE and R2 values, respectively.

5.2. Hyperparameters analysis on model outcome

A deep dive into the hyperparameters of transformer models reveals their profound influence on performance outcomes. This quantitative analysis, focusing on PV power, SI/GHI, Timeseries and input parameters forecasting, highlights how specific configurations and settings impact model efficacy, underscoring the nuanced interplay between computational architecture and forecasting precision.

5.2.1. PV power forecasting models

In the context of PV power prediction, high dimensionality and multiple attention heads appear to be crucial for accurate forecasting. MATNet [85], with its 512-dimensional model and 8 attention heads, achieves MSE of 0.0027, RMSE of 0.0460, and MAE of 0.0245, indicating superior performance. Similarly, Kim's Transformer [208], employing 2 layers each for encoder and decoder and 8 attention heads, achieves PCC of 0.999 and R2 of 0.997, suggesting that higher dimensions [215] and more attention heads [216] can effectively grasp complex solar data patterns.

ISt-LSTM-Informer model [136], combining LSTM units (100 and 120) with an Informer (d_model: 512, n_heads: 6), exhibits RMSE values ranging from 0.617 to 2.285 and MAE values from 0.478 to 0.997. The model showcases how hybrid structures can effectively balance the strengths of LSTM [217] for sequential data and Informer's long-range forecasting capabilities. CNN-LSTM-Transformer [132] integrates CNN with two convolution layers, an LSTM layer of 32 units, and a Transformer model with four encoder and two decoder layers. The values of RMSE = 0.344, MAPE = 0.041 and MAE = 0.393 reflect the model's comprehensive data processing capability, emphasizing the effectiveness of combining CNN's spatial feature extraction [218] with LSTM [217] and Transformer layers for temporal dynamics.

The models of GCTrafo [153] and VIT [120] demonstrate the importance of learning rate adjustments and optimizer choices. GCTrafo [124] has a kernel size [219] of 4 and 8 attention heads, whilst VIT [120] being with a learning rate set at 0.0001 and adjusted based on the epoch number, showing how finely tuned learning rates [220] can optimize model training and output. H-Transformer [144] and Cid et al.'s model [143] have illustrated adaptability. H-Transformer [144] ranges in units from 4 to 256 and learning rates [220] from 0.1 to 0.001, achieving RMSEs of 766 and 1307 on different solar datasets. Cid et al.'s model [143] uses a 24*5-h encoder sequence and 24-h decoder sequence [221], with 2 attention heads and 2 layers each, resulting in an MAE of 0.11827 and RMSE of 0.29098 for power forecasting.

5.2.2. SI/GHI prediction models

In SI or GHI forecasting, the nuanced configuration of hyperparameters within transformer models is critically influential in shaping their performance outcomes. This assertion is bolstered by research findings, such as those presented by Liu et al. [166], which detail models achieving an MAE between 34.21 and 49.53 W/m^2, and AST [87], both of which underscore the efficacy of employing a restrained number of training epochs coupled with early stopping mechanisms[222]. This strategy not only mitigates the risk of overfitting but also underscores the emphasis on precision, thereby bolstering model reliability across a spectrum of forecasting scenarios.

Models like VIT [95], with an nMAPE under 10% and an R2 of 0.85, alongside Informer [83], which reports an MSE of 0.156 and an MAE of 0.262, illuminate the criticality of resource optimization. VIT's [95] implementation across multiple GPUs and a substantial batch size [223], juxtaposed with the computationally demanding architecture of Informer [83], illustrates the pivotal role of computational efficiency in enabling practical, real-time forecasting applications. Furthermore, the CS-TFT [147] model exemplifies adaptability to diverse climatic conditions, suggesting an adeptly calibrated model configuration. In parallel, the Multivariate Transformer [170], with an MSE of 0.3422, reveals a model architecture finely tuned for the nuances of multivariate solar data, striking a harmonious balance between complexity and computational efficiency[224].

The push towards probabilistic forecasting is notably represented in models such as ProSIT [108] and RSAM [109], which utilize distinct attention mechanisms (ProSIT [108]: MSE NL = 0.846, UK = 0.807; RSAM [109]: RMSE for India = 52.45, Hotevilla = 64.87) to adeptly navigate the inherent uncertainties within solar data. This focus marks a growing recognition of the importance of uncertainty quantification [225] in enhancing the robustness of solar energy predictions. Moreover, the RSAM [109] model, with its configuration incorporating multiple attention heads and an expansive feed-forward network [226], underscores the industry's inclination towards models that can intricately parse the complex temporal and spatial dynamics inherent to solar data. Such an analytical approach is indispensable for capturing the nuanced dynamics of SI, thereby significantly elevating the precision of forecasts.

5.2.3. Time series forecasting for solar datasets

Within the domain of time series forecasting for solar energy, an examination of model hyperparameters across various studies reveals patterns and interconnections that are pivotal to their efficacy:

Models such as the Convolutional Transformer [130] and ACT [134] showcase an emphasis on achieving a balance between computational efficiency and model intricacy[56]. Specifically, the Convolutional Transformer [130] is designed with six layers each in its encoder and decoder, alongside a dimensionality[227] set at 64, achieving a correlation coefficient (CORR) of 0.748. This indicates that an equilibrium in the model's layering and dimensional aspects can adeptly handle complex time series datasets without excessive computational burden. In a similar vein, ACT [134], with its structure of three layers in both the encoder and decoder dedicated to short and long-term forecasts, exemplifies an optimal arrangement for managing temporal datasets [228], a key factor for precise SI predictions.

Further, models like AST [87] and PDTrans [105,106] stress the significance of adaptive learning methodologies [220]. AST [87] integrates a discriminator network to elevate sequence-level precision, utilizing the Adam optimizer [229] for concurrent training. This strategy underscores the importance of modulating various loss types to improve forecast accuracy. PDTrans [105,106], adopting an Adam optimizer with a decremental learning rate every two epochs, showcases the efficacy of adaptive learning rate [230] tactics in enhancing training outcomes, especially in probabilistic forecasting scenarios, as reflected by its $\rho 0.5/\rho 0.9$ metrics (the comparative ratio of the 50th to the 90th percentile scores).

The achievements of models like GQFormer [107] and SpringNet [134] highlight the critical role of model structure and hyperparameter refinement. GQFormer [107], despite the absence of specified hyperparameters, registers notable quantile loss figures, implying that a well-balanced model framework can effectively navigate the variability present in solar energy datasets. SpringNet [134], leveraging the Adam optimizer [207] along with Bayesian optimization [231] for hyperparameter refinement, illustrates how meticulous adjustments in dropout rates, layer dimensions [232], and head counts can substantially elevate a model's proficiency in deciphering spatio-temporal patterns within solar energy data.

5.2.4. Models predicting input parameters for solar forecasting

The Informer model, as developed by Jiang et al. [82] and aimed at solar energy forecasting, demonstrates a strategic hyperparameter configuration that bolsters its efficacy and adaptability across different datasets. This model employs a batch-wise parameter update mechanism [233] throughout six epochs of training, reflecting a disciplined learning methodology[230] that mitigates the risk of overfitting. A cornerstone of its training regimen is the utilization of an adaptive learning rate [220], starting at 0.0001 and halving to fine-tune the model's sensitivity to the training dataset. An early stopping feature [222], activated after a patience period of three epochs, prevents excessive training, thus preserving the model's relevance to novel datasets. Additionally, the application of a standard-scaler for data normalization [234], which recalibrates each input feature to a mean of zero and a standard deviation of one, plays a critical role in enhancing the model's operational efficiency. This meticulous arrangement of hyperparameters significantly contributes to the Informer model's distinguished performance and predictive precision in solar energy forecasting.

5.3. Impact of specific hyperparameters

In the nuanced realm of solar energy forecasting, the delineation of specific hyperparameters across diverse models markedly impacts their predictive performance and accuracy. This dynamic is observable through numerous studies that apply a variety of modeling approaches for forecasting SI and PV generation.

5.3.1. Layer configuration (encoder/decoder layers)

The configuration of encoder and decoder layers, along with the allocation of attention heads, plays a crucial role in optimizing the predictive performance and accuracy of models in solar forecasting. This can be illustrated through a detailed examination of various models and their specific architectural choices, highlighting the quantitative effects of these hyperparameters on model metrics.

In the hybrid model developed by Liu et al. [166], which combines Informer and Transformer architectures, the use of 4 encoder layers and 4 decoder layers, each equipped with 8 attention heads, has shown significant improvements. Specifically, increasing the number of encoder layers from 2 to 4 resulted in a 20% decrease in MSE and a 15% decrease in RMSE. This suggests that deeper layers are effective at handling the complex temporal patterns inherent in solar data. Gerges et al. [145] hybrid model, integrating BiLSTM with Transformer layers, features 3 BiLSTM layers and 3 Transformer layers, each with 6 attention heads. By extending the BiLSTM layers to 4, while maintaining 3 Transformer layers, there was a noticeable reduction in MAE by approximately 12%. This improvement indicates that a greater depth in recurrent layers enhances the model's ability to capture short-term dependencies, which is crucial for precise solar forecasting.

Different implementations of Transformer models also demonstrate the impact of layer configurations. For instance, Lara-Benítez et al. [69] use of 6 encoder and 6 decoder layers with 12 attention heads reduced the MAE by around 8% compared to models with fewer layers. Moreover, Mercier et al. [110] adjustment to 16 attention heads, while maintaining the same layer count, further reduced the RMSE by 5%. These outcomes exemplify the significant role of attention mechanisms in improving sensitivity to temporal scales. The ViT [68], adapted for solar forecasting with 12 layers and 16 attention heads, showed a substantial decrease in MSE by 18% relative to standard Transformer models with fewer layers. This model effectively leverages a higher number of layers and attention heads to analyze spatial-temporal data more effectively. Multi-branch ResTrans [101] features 8 encoder layers with a multi-attention mechanism. Increasing the attention heads from 8 to 12 led to a 14% reduction in RMSE and an improvement in the R^2 by 0.05. These enhancements underscore the benefits of additional attention heads in managing the variability and uncertainty in SI data.

Lastly, DualET [235] model utilizes a dual-encoder setup with each encoder having 5 layers and 10 attention heads, exhibited a 10% improvement in model efficiency and a 7% increase in forecast accuracy over models with simpler configurations. This demonstrates how critical the depth and complexity of layer configurations and attention mechanisms are in enhancing the performance of solar forecasting models.

5.3.2. Encoding dimensions

The dimensionality within the encoding phase is of paramount importance. For instance, the Hybrid (Informer-Transformer) model [166] optimizes its encoder and decoder configuration to better capture the complexities of SI. An increase in the dimensionality of these layers typically correlates with improved performance metrics. Specifically, models with higher dimensional encodings have shown to significantly reduce the RMSE and enhance the correlation coefficients, indicating a stronger predictive capability and a better fit to the observed data.

Similarly, Transformer-based models such as the ViT [95] and the Multihead Attention Modified Transformer [162] have been configured with varied dimensional settings across their layers. These settings directly impact the models' efficiency in processing and learning from the solar data. For example, a Vision Transformer [95] with an increased embedding size can more accurately capture the spatial-temporal dynamics, which is crucial for precise solar forecasting. This configuration was noted to decrease RMSE by approximately 17% compared to models with lower dimensionality.

Moreover, ProSIT [108], that incorporates dimensionality adjustments in its encoder to better model the probability distributions of SI show a noteworthy improvement in forecast accuracy. Enhancing the model's dimensionality by 20% was associated with an increase in correlation coefficients by 0.07, underscoring the critical role of dimensionality in modeling uncertainties and variability in solar data.

5.3.3. Learning rate and optimization strategy

The strategic selection of optimization strategies and learning rate adjustments plays a crucial role in enhancing model performance. Detailed analysis of various models reveals that the implementation of the Adam optimizer, known for its efficiency in handling sparse gradients and adaptive learning rate mechanisms, significantly improves forecasting accuracy. For instance, the Hybrid model [166] configuration, utilizes the Adam optimizer with a learning rate initially set at 0.001. Through systematic adjustments to lower this rate to 0.0001 during training phases that exhibited slower convergence, a reduction in Quantile Loss by approximately 20% is reported, underscoring the optimizer's ability to refine the model's response to complex data variations.

Similarly, Gerges et al. model [145], which integrates BLSTM with Transformer layers, also benefits from the Adam optimizer. Gerges highlights an optimization strategy that includes gradient clipping alongside a dynamically adjusted learning rate. This approach not only stabilized the training process but also improved the RMSE by 15% and enhanced the overall correlation coefficient by 0.08 compared to previous iterations without these adjustments. Another notable application is found ViT [95] model, this optimization strategy involved a fine-tuned learning rate decrement from 0.01 to 0.005 over the training epochs, which facilitated a more controlled model fitting. This adjustment led to a significant enhancement in model accuracy, reducing the MAE by 12% and improving the model's efficiency in capturing SI patterns.

5.3.4. Batch size and training epochs

The equilibrium between batch size and training epochs plays a significant role. Achieving this balance is essential in averting overfitting and is imperative for the maintenance of forecast accuracy. Liu et al. [166] hybrid model leverages an optimal batch size of 32 and 100 training epochs. This setup has shown to decrease the RMSE by approximately 10% compared to a baseline model trained with larger batch sizes and fewer epochs. The precise calibration between batch size and epochs in Liu's model underscores the importance of these parameters in refining the model's ability to generalize across diverse solar data scenarios.

Similarly, Gerges et al. [145] model employs a batch size of 64 and 150 training epochs. This configuration facilitates a gradual and thorough learning process, evidenced by a consistent reduction in MAE by about 15% across validation datasets. Gerges et al. [145] approach highlights how a larger batch size, combined with a sufficient number of epochs, can effectively manage the complexities of long sequence data in solar forecasting. ViT model [95], optimized with a smaller batch size of 16 and over 200 epochs, demonstrates an alternative strategy. This model configuration leads to an improvement in the Quantile Loss metric, reducing lower quantile losses by nearly 20%. ViT model illustrates the benefits of smaller batch sizes in providing more frequent updates and adjustments during training, which is crucial for capturing subtle nuances in SI data.

5.3.5. Adversarial and probabilistic elements

The ACT model [86] incorporates adversarial training elements to test the model's resilience against perturbations in solar data. This model demonstrated a reduction in Quantile Loss, with the 50th and 90th quantiles improving by approximately 18% and 25%, respectively, compared to models without adversarial components. This improvement underscores the effectiveness of adversarial training in enhancing the model's ability to generalize across various conditions, leading to more reliable forecasts.

Another notable model, the PDTrans [106], leverages probabilistic forecasting methods to estimate the range and likelihood of potential SI values. This model has shown a significant increase in the accuracy of forecasting, with a decrease in RMSE by around 22% compared to non-probabilistic models. The incorporation of probabilistic elements enables the model to effectively handle the inherent uncertainties in solar data, providing a more nuanced understanding of potential variability in the forecasts. Similarly, AST Model [87] employs sparse adversarial training techniques to streamline the model's focus on the most impactful features within the data. This approach has not only improved the model's efficiency by reducing the computational load but also enhanced the prediction accuracy, as evidenced by a 20% improvement in MAE.

5.3.6. Discriminator networks and sequence-level accuracy

Employing discriminator networks to bolster sequence-level accuracy substantially influences model outcomes. These networks facilitate a comprehensive regularization, ensuring predictions are closely aligned with actual temporal patterns, thus improving forecast reliability and precision [236]. The ACT [86] employs a discriminator network designed to distinguish between actual and predicted SI sequences. This model, through adversarial training, has demonstrated a marked improvement in forecasting accuracy, with the discriminator enhancing the generator's ability to produce realistic irradiance outputs. The ACT model recorded a 12% decrease in RMSE and a 15% increase in sequence-level accuracy compared to conventional models, illustrating the critical role of discriminator networks in refining predictive performance.

Additionally, the AST [87] utilizes a sparse representation in conjunction with a discriminator to focus the model's learning on the most salient features of the solar data. This approach not only streamlines the model's computational demands but also significantly boosts accuracy. The model achieved a 10% improvement in MAE and demonstrated superior sequence-level forecasting capabilities, particularly in capturing the temporal dynamics of SI.

5.3.7. Hyperparameter search and learning rate scheduling

Engaging in hyperparameter optimization and learning rate [220] scheduling is fundamental for attaining peak performance. Techniques such as Bayesian optimization [134] and strategic learning rate adjustments emphasize the critical nature of parameter refinement, as evidenced by improved MAE and RMSE metrics, showcasing the significance of a thorough approach in parameter tuning for superior forecasting capabilities [237]. For instance, the MATNet model [85], utilizes an input embedding dimensionality $dmodel = 512$, with specific attention to learning rate adjustments. This model employs a dynamic learning rate schedule, starting with a rate of 0.01 and reducing it by 10% every 50 epochs based on performance metrics. This strategy led to a noticeable improvement in forecast accuracy, reducing the MSE by 25% compared to a fixed learning rate strategy.

Another compelling example is Liu et al. [166] model, which incorporates a comprehensive hyperparameter search that included varying the number of encoder and decoder layers and tuning the model size. This approach involves using a grid search to systematically test combinations, resulted in selecting an optimal configuration that achieved a 20% lower RMSE than the initial settings. Moreover, the ACT [86] integrates an advanced learning rate scheduler. This model begins with a higher learning rate to promote rapid convergence early in training and gradually decreases the rate to fine-tune the model's weights as training progresses. This method enhanced the model's ability to adapt to varying solar data dynamics, improving its sequence prediction accuracy by approximately 15%.

In summary, meticulous hyperparameter tuning culminates in notable advancements in the accuracy and precision of models designed for solar time series forecasting. This precision is indispensable for scenarios necessitating exact forecasts of solar energy production. Effective hyperparameter configurations not only ensure model accuracy but also computational efficiency, vital for real-time forecasting. Moreover, models refined through precise hyperparameter tuning demonstrate enhanced robustness and adaptability across various solar datasets and temporal frames. The strategic inclusion of adversarial and probabilistic elements, driven by hyperparameter adjustments, further elevates the model's proficiency in managing the inherent variability within solar data forecasting, rendering them more dependable and functional for practical use.

6. Comprehensive analysis and path forward

6.1. Advancements and synergies in solar energy forecasting

The evolution of solar forecasting models, particularly through the lens of transformer-based architectures and their synergy with other computational techniques such as CNNs and RNNs, offers a deep dive into the sector's strides towards addressing the intricate demands of solar energy data interpretation and prediction. The gradual shift from traditional transformer models to more nuanced hybrid and specialized variants [130,133] reflects a deliberate adaptation to the layered complexities inherent in forecasting solar energy outputs. This integration [128,129] has been pivotal, significantly enhancing the capacity of these models to dissect and understand the multifaceted temporal and spatial dependencies that characterize solar energy data. The advancement is further evidenced by the deployment of sophisticated attention mechanisms and innovative encoder-decoder configurations [76,100,101], marking a substantial leap in forecasting precision and efficiency.

Customization emerges as a notable trend, with models such as ACT [86] and ResTrans [100] being specifically designed to confront the unique challenges posed by solar forecasting. This strategic focus on tailor-made solutions underlines a broader shift towards models that are finely attuned to the peculiarities of solar data, including its environmental variabilities and the cyclical nature of SI.

The exploration into the temporal dynamics of forecasting unveils a clear preference for short-term forecasting [72,95,166,170,171], driven by the operational imperatives of energy systems. However, the relative neglect of medium-term forecasting [111,112] suggests an untapped potential for research endeavors aimed at finding a harmonious balance between forecast horizon, accuracy, and computational demands. In the domain of very short-term forecasting, the fusion of vision-based and transformer technologies [166] has shown immense promise. Nonetheless, the diversity in metrics used for model comparison [95,166,171] points to a pressing need for standardized evaluation frameworks that would allow for more direct and comprehensive assessments of model performance. Short-term forecasting has notably benefited from the prowess of transformer models, especially in the context of SI/GHI prediction [108,147], with the methodological inclusion of environmental factors [83] marking a significant step forward. Yet, the quest for universally applicable models necessitates further explorations beyond the current paradigm. The inquiry into medium-term forecasting, as spearheaded by Sinha et al. [111,112], paves the way for new research avenues, particularly in the realms of sub seasonal and week-ahead forecasts. This not only suggests a fertile ground for future exploration but also emphasizes the potential of transformer models in enhancing probabilistic forecasting capabilities. The ambiguity surrounding forecasting times across different studies [78,119,130] poses a challenge to the practical application of these models, highlighting a critical area for future research to demystify the operational time frames of transformer models and ensure their findings are tangibly applicable to real-world forecasting needs.

The pursuit of computational efficiency [10,201] illuminates a crucial balancing act in solar forecasting: the intersection of model accuracy with the pragmatics of model deployment. While Transformer models [66,67,77] have led the charge in advancing data processing capabilities, their computational demands necessitate a judicious approach to model selection and application [202].

The impact of hyperparameters on forecasting accuracy cannot be overstated. The dimensionality of models and the intricacy of attention mechanisms [85,120], alongside the choice of learning rates and optimization strategies [83,120], play determinative roles in model outcomes. The strategic configuration of encoder-decoder layers [208] and the meticulous adjustment of batch sizes and training epochs [132] are crucial for attaining peak forecasting performance.

6.2. Navigating the hurdles: Challenges to overcome

The exploration into the advanced architectures of solar forecasting models, particularly those employing transformer-based designs and their integrations, has unveiled several pivotal limitations that stand to impact the field's progress and the practical application of these models.

A primary limitation observed is the significant reliance on comprehensive, high-quality datasets [79], which are not universally available. This dependency not only challenges the models' applicability across various regions, particularly those with sparse data availability but also raises concerns regarding the global utility and scalability of these forecasting solutions. The specificity of datasets like SURFRAD [111,112] further compounds this issue, potentially restricting model generalization across diverse geographical and climatic conditions and questioning the broader applicability of findings within the global context of solar forecasting.

Moreover, the computational demands of hybrid and specialized transformer models [69,137] pose substantial challenges for real-time forecasting applications, especially in scenarios where speed and efficiency are paramount. This issue is exacerbated by the high resource demands of models excelling in very short-term forecasting [95,166], which may restrict their use in resource-constrained environments. Furthermore, the underrepresentation of medium-term forecasting [79] highlights a critical gap in current research efforts. Despite its importance for effective energy planning and management, medium-term forecasting has been largely overlooked, suggesting a research trajectory that needs addressing. This oversight is attributed to the difficulties in balancing computational efficiency with the necessary accuracy for medium-range predictions, underscoring a broader trend of forecast horizon bias. The prevailing focus on short-term forecasting [72,170,171] overshadows the need for medium to long-term forecasting solutions, crucial for strategic planning within energy systems.

The resource intensiveness [10,238] and increasing sophistication of models [86,148] add layers of complexity in tuning, maintaining, and interpreting these systems, potentially deterring their adoption in practical applications. Furthermore, the intricate process of hyperparameter tuning and its impact on computational efficiency presents significant challenges. The quest for optimal hyperparameter configurations, while aiming to enhance model performance, raises concerns about the practical deployment of these optimized models in scenarios with limited computational resources.

These challenges collectively underscore the need for a balanced approach that considers the practical limitations of current solar forecasting models. Addressing these limitations necessitates a concerted effort towards developing models that not only exhibit high accuracy and efficiency but are also adaptable, scalable, and capable of being deployed within a wide range of operational contexts.

6.3. Blueprint for innovation: Enhancing precision and efficiency in solar forecasting

In propelling future advancements in solar forecasting, it is crucial not only to focus on refining transformer models that strike a delicate balance between computational efficiency and forecast accuracy but also to confront the significant challenges outlined in the section of limitations. This necessitates a dedicated emphasis on pioneering innovations in model architecture aimed at curtailing computational burdens without compromising performance. Key strategies include optimizing attention mechanisms and enhancing data processing techniques [87,90], crucial steps towards this objective. Furthermore, the exploration of hybrid models that amalgamate the robust attributes of transformers with other computational methodologies [128,129,144] holds immense potential to bolster adaptability and tackle the intricate nuances inherent in solar forecasting, particularly in areas characterized by scant data or diverse climatic dynamics.

Addressing the considerable gap in medium-term solar forecasting [111,112] represents a critical priority. Directed research endeavors should be geared towards devising models that strike an optimal balance between computational efficiency and predictive accuracy for medium-range forecasting. This entails delving into inventive model frameworks precisely tailored to deftly manage the challenges specific to medium-term prediction. Leveraging approaches like cross-domain applications and transfer learning [77,80] presents a promising avenue to alleviate data scarcity concerns and potentially uplift the efficacy of medium-term forecasting models, thereby expanding their utility and impact.

Furthermore, future endeavors should extend to optimizing both transformer and hybrid models for use in resource-constrained environments. Developing lightweight yet accurate models and investigating novel hybrid frameworks that combine various computational strategies without significantly increasing complexity could revolutionize solar forecasting practices. Establishing benchmark datasets and standardizing performance metrics across models will play a critical role in facilitating clear comparisons of model efficiency and efficacy, aiding in the precise selection of models for specific forecasting tasks.

Within the domain of hyperparameter optimization, the development of automated techniques such as Bayesian optimization or reinforcement learning holds the promise of expediting the model development process substantially, thereby alleviating the computational load. Probing into the implications of emerging hyperparameters on the resilience and transferability of models across varied datasets and forecasting scenarios assumes paramount importance. Engaging in cross-validation studies to evaluate the generalizability of optimized hyperparameters across diverse geographical locations and temporal scales will furnish invaluable insights into their versatility and efficacy, setting the stage for the development of more accurate and globally applicable solar forecasting models.

7. Conclusion

This comprehensive review has provided a critical exploration of the evolving landscape of solar energy forecasting, with a particular focus on the integration of transformer models alongside computational techniques such as CNNs and RNNs. Below we summarize key advancements, pinpoint significant gaps in the existing literature, and propose clear pathways for future research.

(1) The shift towards hybrid and specialized transformer models has significantly enhanced the accuracy of solar forecasting, with these integrations improving models' ability to analyze spatial-temporal complexities. Noteworthy advancements include custom solutions like the ACT and ResTrans models, which demonstrate tailored approaches to the unique dynamics of solar data. However, there exists a noticeable bias towards short-term forecasting, thus underlining the need for more exploration in medium-term forecasting, crucial for effective energy management and planning.
(2) An evident gap in current research is the absence of a unified framework for evaluating and comparing model performance. Future endeavors should concentrate on establishing benchmark datasets and standardizing metrics to enable more coherent comparisons across different models and studies.
(3) The limited focus on medium-term forecasting emphasizes a significant research void. Subsequent studies should concentrate on developing models that effectively balance computational efficiency with the accuracy required for medium-range forecasts. Exploring models with streamlined attention mechanisms or implementing techniques such as transfer learning could offer viable solutions to address this gap.
(4) Despite the advancements in forecasting precision, the substantial computational demands of current models emphasize the need for optimization. Future initiatives should prioritize the development of models that ensure computational efficiency without compromising forecasting accuracy, potentially through innovations in model architecture or hybrid frameworks that intelligently combine computational strategies.
(5) The crucial role of hyperparameters in forecasting performance highlights the importance of advanced optimization techniques. Streamlining the tuning process and investigating the effects of new hyperparameter configurations are fundamental steps towards enhancing the efficiency and accuracy of models.

In conclusion, this review has conducted a comprehensive analysis of solar forecasting through transformer models, outlining significant advancements and critical limitations. While notable progress has been achieved, particularly in customizing model architectures and strategically integrating computational techniques, substantial challenges remain. It is imperative to address these challenges, especially regarding medium-term forecasting and computational efficiency, to advance the field. Future research efforts should prioritize developing models that not only demonstrate high precision and efficiency but also exhibit adaptability and scalability across various operational contexts. Emphasizing hyperparameter optimization and exploring automated tuning methods will be pivotal in attaining these objectives, ensuring that solar forecasting models continue to evolve to meet the ever-changing demands of renewable energy management and grid integration.

CRediT authorship contribution statement

M.F. Hanif: Writing – original draft, Resources, Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization. **J. Mi:** Writing – review & editing, Supervision, Methodology, Conceptualization, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgement

The financial support of China Scholarship Council (CSC) for M.F. Hanif's PhD program is acknowledged.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apenergy.2024.123541.

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